

Coordinating Development Under Political Risk^{*}

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Abstract

Political risk and coordination failure are major barriers to investment and growth in developing countries. We show that political risk can itself induce coordination failure and propose a subsidy scheme to mitigate miscoordination. The program guarantees investors a minimum return while clawing back returns above a threshold. By leveraging strategic complementarities, the clawback screens out investors who would have invested absent subsidies, while the guarantee attracts those deterred by political risk and strategic uncertainty. Program costs per unit of capital are single-peaked in available capital for investment, implying economies of scale in subsidy programs.

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1 Introduction

There is a vast and growing global investment gap in achieving sustainable development in low- and middle-income countries.¹ The international community devotes enormous resources to confronting this challenge.² How can these resources be most effectively leveraged to mitigate the investment gap and promote sustainable growth?

The answer depends on understanding why investment³ remains low in many developing countries. Political risk and coordination failure are two leading explanations ([World Bank, 2017, 2024](#)). Conflict, policy uncertainty, expropriation and confiscatory taxation, corruption, and weak rule of law are associated with low economic progress ([Acemoglu, Johnson, and Robinson, 2001](#); [Alfaro, Kalemli-Ozcan, and Volosovych, 2008](#); [Boix, 2003](#); [Caldara and Iacoviello, 2022](#); [Jensen, 2008](#); [North and Weingast, 1989](#); [Reinhart and Rogoff, 2004](#)).⁴ Similarly, countries may be trapped into low investment and growth if investors fail to coordinate and internalize positive externalities in demand, infrastructure, and technology adoption ([Buera et al., 2021](#); [Choi and Shim, 2024](#); [Garg, 2025](#); [Murphy, Shleifer, and Vishny, 1989](#); [Rodríguez-Clare, 2005](#); [Rodrik, 1996](#)). As the 2017 World Development Report emphasizes “coordination is required to induce investment and innovation. Both depend on firms and individuals believing that others will also invest” ([World Bank, 2017](#), p. 56).

Our key insight is that political risk induces coordination failure among investors, but well-designed subsidy programs can nearly eliminate miscoordination at low cost.⁵ Political risk and coordination failure are intertwined: A larger number of investors reduces political risk,

¹According to the *World Investment Report 2023*, “the investment gap across all [Sustainable Development Goal] sectors has increased from \$2.5 trillion – estimated in [*World Investment Report 2014*]... to more than \$4 trillion per year today... The increase is the result of both underinvestment and additional needs” ([United Nations Conference on Trade and Development, 2023](#), p. xv).

²For example, the World Bank Group alone committed \$117.5 billion towards development projects in the fiscal year 2024 ([Multilateral Investment Guarantee Agency, 2024](#), p. 4).

³We use a broad notion of investment, including physical and human capital, technology, and innovation.

⁴The literature is vast. See [Abadie and Gardeazabal \(2003\)](#), [Alesina and Perotti \(1996\)](#), and [Cerra and Saxena \(2008\)](#) for unrest and conflict; [Baker, Bloom, and Davis \(2016\)](#) and [Fernández-Villaverde et al. \(2015\)](#) for policy uncertainty; [Boehm and Oberfield \(2020\)](#) and [Ponticelli and Alencar \(2016\)](#) for weak rule of law; and [Fisman and Svensson \(2007\)](#) and [Mauro \(1995\)](#) for corruption.

⁵Political risk can induce other forms of coordination failure, e.g., between lenders and borrower-investors ([Chang, 2010](#)). Our approach applies to such settings as long as subsidies can be conditional on investments.

which in turn encourages further investment. For example, [Shadmehr \(2019\)](#) argues that when technology exhibits complementarity between capital and labor, greater investment raises wages, thereby reducing the likelihood of a successful revolution. Similarly, greater investment may yield a more “business-friendly” median voter and encourage more far-sighted policies, reflecting both improved economic conditions and more promising future opportunities ([Bernhardt, Krasa, and Shadmehr, 2022](#); [Chang, 2010](#)). Strategic uncertainty, combined with politically induced complementarity, leads to inefficiently low investment. If a potential investor was sure that many others would invest, it would anticipate a lower political risk, and become more optimistic about returns. However, investors remain uncertain about each other’s assessments of investment prospects and therefore decisions, leading to miscoordination. If, instead of many, there was one large potential investor, there would be no miscoordination inefficiencies. However, given the vast scales, resolving the investment gap is not feasible for one or a few entities.⁶ Addressing the investment gap therefore requires the participation of many private investors.

Building on the insights of [Shen and Zou \(2024\)](#), we propose a subsidy program⁷ that nearly eliminates miscoordination at a lower cost than natural alternatives. The program features voluntary investor participation, guarantees a minimum return, claws back profits exceeding a specified maximum, and compensates pre-guarantee losses up to the maximum feasible rate. We call this the optimal *guaranteed return with profit- and loss-sharing* (*GPLS*) program. We show that loss-sharing is essential: without it, any *guaranteed return with profit-sharing* (*GPS*) program can be significantly more expensive—while still better than optimal guarantees without profit-sharing. A restricted version of GPLS (*R-GPLS*), in which the profit- and loss-sharing rates are constrained to be equal, is also less cost-effective. We refer to this version as the Shen–Zou (SZ) program.⁸

⁶Even if it were feasible, such a centralized setting would introduce additional frictions, including those associated with monopolies—[Lucas \(1990\)](#) identified monopolies, supported by imperial powers, as a key political risk.

⁷One may view the program as an insurance program. We call it a subsidy program to emphasize that underwriters will transfer funds to participants in expectation.

⁸We focus on applying recent theoretical advances to address the global investment gap. Methodologically, the core analysis of [Shen and Zou \(2024\)](#) is developed in a regime-change setting, whereas our framework

Development agencies can, in principle, induce any level of investment by offering sufficiently high payoffs to swamp all risk. However, such guarantees are prohibitively expensive, and we therefore maintain the realistic assumption that it is infeasible for development agencies to offer programs that induce investment even in the worst-case scenarios. Even when an agency can elicit a desired aggregate investment level, it is not obvious that it should do so by subsidizing investment through guaranteed payoffs. Any program that offers a guaranteed return will attract not only investors who would not invest without the program but also those who would have invested anyway, substantially increasing costs.

The key is to design an effective screening mechanism. One might imagine asking investors directly and offering subsidies only to those who honestly reveal that they would not invest without them. Yet profit-maximizing investors cannot be relied upon to forgo profit for honesty, and it is infeasible to identify which investors would have invested absent subsidies. To screen out these investors, the optimal program claws back any profits exceeding a maximum return when investments succeed. Those sufficiently optimistic to invest without the program then choose not to participate, preferring to retain the additional profits. This defines the optimal GPS program.

The optimal subsidy has a distinct structure. Investors who opt in receive a minimum guaranteed return on their investment, but their payoffs are capped, as the program claims all returns above the maximum. The guarantee attracts investors who would otherwise not invest, while the cap screens out those who would have invested without the program.

The optimal GPS program improves upon a simple guaranteed-payoff scheme, but its screening can be refined further. If the agency knew each investor’s degree of optimism about investment prospects, it would provide just enough funds to induce the desired investment—offering no incentives to those who would have invested anyway and continuing toward more pessimistic ones, who require larger subsidies, until the target level is reached. However,

features continuous net payoffs. Their Appendix B outlines an extension that encompasses R-GPLS. Our theoretical contribution lies in analyzing both more and less restrictive interventions than their proposed program (GPS and GPLS), showing that the optimal GPLS program is more cost-effective—see Sections 5.1 and 6—and establishing economies of scale in these subsidy programs—see [Proposition 6](#).

the agency does not observe investors' degrees of optimism, and more optimistic investors have incentives to mimic pessimistic ones to obtain more subsidy (Börger, 2015). Consequently, under the optimal GPS program, nearly all participants receive larger subsidies than they would if the agency knew their types, leaving scope for further improvement.

The optimal GPLS program addresses this problem by adopting a more flexible structure. Rather than offering a large guaranteed return to all participants, it provides a smaller guarantee while sharing a portion of investors' pre-guarantee losses. This design makes participation less attractive to less pessimistic investors, who expect smaller benefits from loss sharing. These new investors now invest without joining the program, anticipating that it will induce more pessimistic investors to participate and invest. This remarkable feature illustrates how the optimal GPLS program leverages coordination incentives to expand investment at minimal cost.

We further show that the optimal programs (GPS and GPLS) exhibit increasing returns to scale: the expected cost per unit of capital is strictly single-peaked in the available capital stock for investment. This implies that development agencies may benefit from setting sufficiently ambitious investment targets. Indeed, once the available capital stock surpasses a threshold, eliminating miscoordination at higher levels of investment can become paradoxically cheaper. Intuitively, the programs pay potential investors to behave as if all other investors will invest. A larger capital stock generates two competing forces. First, strategic uncertainty weakens the marginal effect of capital stock on reducing political risk, because only a fraction of available capital is invested under strategic uncertainty. This raises program costs. Second, the marginal effect of investment on political risk diminishes. This reduces program costs. When capital stock is low, the first effect dominates; as capital stock grows, the second effect dominates.

Our approach complements the solutions offered in the political risk literature to foster investment and growth, including trade agreements (Büthe and Milner, 2008, 2014), institutional change to introduce checks and balances (Acemoglu, Naidu, et al., 2019; Ferraz

and Finan, 2011; Gehlbach and Keefer, 2012; La Porta et al., 2004; North, 1990) and legal protection for property rights (Acemoglu and Johnson, 2005; Besley and Ghatak, 2010; Li and Resnick, 2003), as well as transparency, monitoring, and deliberation (Banerjee, Duflo, et al., 2020; Djankov et al., 2010; Ferraz and Finan, 2008; Fujiwara and Wantchekon, 2013; López-Moctezuma et al., 2022; Olken, 2007). It parallels recent microfinance literature on the importance of contract structure and borrower heterogeneity, highlighting the need for better screening to foster development (Balboni et al., 2022; Banerjee, Breza, et al., 2019; Bari et al., 2024; Bryan, Karlan, and Osman, 2024).

The growth literature on coordination failure among investors suggests subsidies (Buera et al., 2021; Rodrik, 1996), highlighting that income-based (as opposed to our investment-based) incentive programs are ineffective (Bond and Pande, 2007), but does not focus on identifying efficient programs. Given the vast scale involved in international development, designing optimal programs that achieve the same outcome at lower costs is crucial. We build on recent developments in literature on screening in coordination games (Morris and Shadmehr, 2023; Shen and Zou, 2024) to identify such programs, comparing their costs with natural sub-optimal alternatives.

A broader literature studies contracting and subsidies in the presence of complementarity (Halac, 2025), including in global games (Luo and Yang, 2023). However, its focus is not on screening, e.g., Sákovics and Steiner (2012) analyze optimal subsidies to ex-ante heterogeneous agents based on observable characteristics. Similarly, aside from Morris and Shadmehr (2023), the literature on coordination in conflict does not involve screening (Boix and Svolik, 2013; Bueno de Mesquita and Shadmehr, 2023; Chau, Hassan, and Little, 2025; Chen and Suen, 2017, 2021; Correa, 2025; Correa, Nandong, and Shadmehr, 2025; Egorov and Sonin, 2021; Gieczewski and Kocak, forthcoming; Little, 2012, 2016; Tyson and Smith, 2018). See Egorov and Sonin (2024) and Cantoni et al. (2024) for reviews.

The international community recognizes political risk as a key obstacle to investment in developing countries (United Nations Conference on Trade and Development, 2025; World

[Economic Forum, 2024](#)). Development agencies—such as the World Bank Group and the U.S. International Development Finance Corporation—offer various instruments to mitigate political risk and encourage private investment. These include guarantees that cover losses due to political incidents (e.g., political risk insurance), subsidies (e.g., below-market loans), and blended finance, which combines public and private capital through multiple instruments ([Arel-Bundock, Peinhardt, and Pond, 2020](#); [Multilateral Investment Guarantee Agency, 2023](#)).⁹ In [Section 6.1](#) we examine a sample of these instruments and show how their design compares to the optimal Guaranteed Return with Profit- and Loss-Sharing (GPLS) program. This comparison highlights how widely used programs align with—or diverge from—an approach explicitly aimed at eliminating coordination failure.

2 Model

There is a continuum 1 of investors, indexed by $i \in [0, 1]$, each endowed with $\bar{K}$ units of capital. They simultaneously decide whether to invest in the global market or in a developing country. The expected return on capital in the global market is r . The return from investing in the country depends on productivity and the resolution of political risk.

We model political risk as an effective tax rate $T \in [0, 1]$. This effective tax reflects not only formal taxation but also the likelihood of expropriation, destruction from conflict, corruption costs ([Shleifer and Vishny, 1993](#)), and weak rule of law and contract enforcement.

Let $k_j \in \{0, 1\}$ denote an investor j 's decision, where 0 means investing in the global market and 1 means investing in the developing country, so that the aggregate invested capital is $K = \bar{K} \int k_j dj$. The production technology is $Y = A(\underline{K} + K)$ where $A > 0$ is the total factor productivity, and $\underline{K} > 0$ is an exogenous, immobile capital in the country, e.g., land ([Boix, 2003](#); [Shadmehr, 2019](#)). A share $\alpha \in (0, 1)$ of the output is divided equally among

⁹See <https://ida.worldbank.org/en/financing/ida-private-sector-window> for information on the International Development Association's Private Sector Window and <https://www.ifc.org/en/what-we-do/sector-expertise/blended-finance/how-blended-finance-works> for the International Finance Corporation's Blended Finance.

the investors. An investor's payoff from investing in the country is the after-tax return on her capital: $(1 - T)\alpha A\bar{K}$.

Empirical studies emphasize that greater investment and better economic conditions tend to reduce political risk (Bazzi and Blattman, 2014; Blattman and Annan, 2016; Blattman and Miguel, 2010; Brückner and Ciccone, 2010; Caldara and Iacoviello, 2022; Johns and Wellhausen, 2016; Malesky, 2009). In our model, T captures political risk. To be consistent with this empirical regularity, we could directly assume that T decreases in output Y , and hence in aggregate capital K . Instead, we provide a micro-foundation for this relationship, which we use throughout the paper. Specifically, after investment and production occur, the representative citizen in the country receives a share $(1 - \alpha)$ of output and chooses the tax rate T . The citizen's payoff is

$$u_c = (1 - T)(1 - \alpha)Y + 2\sqrt{\delta TY},$$

where TY is the tax revenue and δ captures the citizen's preference for taxation, encompassing the productivity of tax revenues, ideological attitude, or other considerations.¹⁰

The productivity A in the developing country is uncertain. Investors share a prior that $A \sim U[\underline{A}, \bar{A}]$. Each investor i receives a noisy private signal x_i about A , where $x_i = A + \sigma\epsilon_i$, $\sigma > 0$, with $\epsilon_i \sim iid U[-1, 1]$. The fundamental A and noise ϵ_i s are independent—see, e.g., Angeletos and Lian (2016).

The game proceeds as follows. First, nature determines A and ϵ_i s, and investors observe their signals, x_i s. Next, investors simultaneously make investment decisions $(k_i)_{i \in [0, 1]}$. Then, production takes place and players receive their shares. Then, the effective tax rate T is chosen. Payoffs are received and the game ends.

¹⁰An interpretation of the representative citizen is the median voter in a setting with a continuum 1 of workers, each providing 1 unit of labor inelastically, receiving the labor share $(1 - \alpha)$ equally, and then voting on a tax rate T that finances the public good $g = TY$. In other settings, the representative citizen may instead be interpreted as the government, or a fictitious agent encompassing the government, citizens, and rebels. Our micro-foundation in the text should be viewed as a modeling tool. What matters for our results is that, in equilibrium, T will be decreasing in output Y , and hence in aggregate capital K —see Lemma 1.

An investor i 's strategy $\sigma_i : [\underline{A} - \sigma, \bar{A} + \sigma] \rightarrow \{0, 1\}$ is a mapping from her signal x_i to her investment decision k_i . The representative citizen's strategy $\sigma_c : \mathbb{R}_+ \rightarrow [0, 1]$ is a mapping from the output to an effective tax rate $T \in [0, 1]$. The equilibrium concept is Perfect Bayesian Nash equilibrium. We focus on symmetric monotone strategies in which an investor i invests if and only if i 's signal x_i is above a threshold x^* .

3 Inefficient Investment and Strategic Uncertainty

We begin by analyzing the citizen's choice of effective tax rate. The citizen's problem is:

$$\max_{T \in [0, 1]} (1 - T)(1 - \alpha)Y + 2\sqrt{\delta(TY)} \quad (1)$$

Lemma 1. *The equilibrium level of effective tax rate is*

$$T^* = \min \left\{ \frac{\delta}{(1 - \alpha)^2} \frac{1}{Y}, 1 \right\} = \min \left\{ \frac{\delta}{(1 - \alpha)^2} \frac{1}{A(\underline{K} + K)}, 1 \right\}. \quad (2)$$

All omitted proofs are in Online Appendix B. The tax rate is decreasing in output, and hence in the total factor productivity A and in the aggregate investment K . This generates a force for strategic complementarity among the investors.

For a given aggregate capital, the tax rate is a random variable because productivity A is random. When more investors invest in the country, this reduces the tax rate in the FoSD sense: the distribution of tax rate shifts to left, raising an investor's incentives to also invest in the country.

We now consider the investors' decisions. Given productivity A , aggregate capital K , and tax rate T^* , an investor's net rate of return from investing in the country versus in the global market is

$$\alpha \pi(K, A) := (1 - T^*)\alpha A - r = \alpha A - \frac{\alpha \delta}{(1 - \alpha)^2} \frac{1}{\underline{K} + K} - r, \quad (3)$$

where we substituted T^* from [Lemma 1](#), assuming that T^* is interior, and recognizing that

investors anticipate the tax rate for any given productivity A and aggregate capital K .

To ease exposition and have unique equilibrium, we maintain the following assumption.

Assumption 1. Let $A_l = \underline{A} - \pi(\bar{K}, \underline{A}) = \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2(\underline{K} + \bar{K})}$ and $A_h = \bar{A} - \pi(0, \bar{A}) = \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2 \underline{K}}$.

1. $\underline{A} > \frac{\delta}{(1-\alpha)^2 \underline{K}}$
2. $\underline{A} < A_l - \sigma$
3. $\bar{A} > A_h + \sigma$

Part 1 ensures that the tax rate is always less than 1. Parts 2 and 3 ensure the existence of lower and upper dominance regions needed for uniqueness. To ease exposition, when indifferent, we assume that a potential investor will invest in the country.

Proposition 1 (Complete Information Benchmark). *Suppose productivity A is known.*

- If $A < A_l$, then there is a unique equilibrium in which no one invests.
- If $A \geq A_h$, then there is a unique equilibrium in which everyone invests.
- If $A \in [A_l, A_h)$, then there are multiple equilibria.

When the productivity is sufficiently large, $A > A_h$, the rate of return is higher in the country than in the global market even if the aggregate investment is negligible. Then, there is a unique equilibrium in which all potential investors invest in the country. Similarly, when the productivity is sufficiently small, $A < A_l$, the rate of return is lower in the country even if all potential investors invest in the country. Then, there is a unique equilibrium in which no potential investor invests in the country. In between, there are multiple equilibria, including one in which all invest, and one in which no one invests.

We now turn to the incomplete information setting with uncertain productivity and information asymmetry. Investors use their information to assess the expected net payoff from investing. They have uncertainty about the productivity and they have strategic uncertainty about other investors' behavior. For a given realization of productivity A and other investors'

cutoff x^* for investing, the aggregate invested capital in the country is

$$K(A; x^*) = \bar{K} \Pr(x_i \geq x^* | A), \quad (4)$$

which is increasing in productivity. Thus, higher productivity directly increases incentives to invest by raising the expected pre-tax returns, and it indirectly increases incentives by reducing the effective tax and hence the expected after-tax returns. It does so because higher productivity reduces the tax rate by raising the output, both directly through the productivity channel A and indirectly through increasing the aggregate capital $\underline{K} + K(A; x^*)$.

An investor i with signal x_i thus estimates i 's net expected payoff $\mathbb{E}[\alpha\pi(K(A; x^*), A) | x_i]$, investing if and only if $x_i \geq x^*$. The investor must be indifferent between investing in the country and in the global market at the threshold signal $x_i = x^*$. It follows that any x^* that satisfies $\mathbb{E}[\pi(K(A; x^*), A) | x_i = x^*] = 0$ constitutes an equilibrium. Obviously, if a solution exists, we must have $x^* \in [A_l, A_h]$: an investor with signal $x_i > A_h$ has a dominant strategy to invest in the country; and one with signal $x_i < A_l$ has a dominant strategy not to.¹¹ To assess her net expected payoff from investing, $\mathbb{E}[\alpha\pi(K(A; x^*), A) | x_i]$, the marginal investor with the threshold signal $x_i = x^*$ must estimate the tax rate and, hence, aggregate investment $K(A; x^*)$. A key observation is that when $\sigma < \min\{\bar{A} - A_h, A_l - \underline{A}\}$, so that $x^* \pm \sigma$ is away from the boundaries, we have:

$$\Pr(x_j \geq x^* | A) | x_i = x^* \sim U[0, 1], \quad (5)$$

so that the marginal investor always believes that the aggregate (new) capital in the country, $K(A; x^*)$, is uniformly distributed on $[0, \bar{K}]$. Then, the indifference condition becomes

$$\alpha \mathbb{E}[A | x_i = x^*] - \alpha \mathbb{E}[A \cdot T^* | x_i = x^*] - r = \alpha x^* - \int_0^{\bar{K}} \frac{\alpha \delta}{(1 - \alpha)^2} \frac{1}{\bar{K}} \frac{dK}{\underline{K} + K} - r = 0, \quad (6)$$

¹¹When no one else invests ($K = 0$), i invests whenever her private signal is greater than A_h , because $\mathbb{E}[\pi(0, A) | x_i = A_h] = 0$. Similarly, when everyone else invests ($K = \bar{K}$), i does not invest if her private signal is below A_l , because $\mathbb{E}[\pi(\bar{K}, A) | x_i = A_l] = 0$.

yielding a unique solution. In fact, it is straightforward to confirm that the setting satisfies the standard assumptions for the common value setting in [Morris and Shin \(2003\)](#). The above arguments adopt their proof of Proposition 2.2 to the setting with uniform distributions.¹²

Proposition 2 (No Intervention Benchmark). *Let $\bar{\sigma} = \min\{\bar{A} - A_h, A_l - \underline{A}\}$. If $\sigma < \bar{\sigma}$, there is a unique symmetric equilibrium in cutoff strategies in which an investor i with signal x_i invests in the country if and only if $x_i \geq x^*$. Moreover, x^* is the unique solution to $\int_0^{\bar{K}} \pi(K, x^*) dK = 0$:*

$$x^* = \frac{r}{\alpha} + \frac{\delta}{(1 - \alpha)^2} \frac{1}{\bar{K}} \log \left(\frac{\bar{K} + \underline{K}}{\underline{K}} \right).$$

Comparative statics are intuitive.

Corollary 1. *Higher global market returns r and preference for taxation δ reduce investment incentives, while higher immobile capital $\underline{K}$ and available investment $\bar{K}$ raise investment incentives—because they reduce taxes: $\frac{dx^*}{d\bar{K}}, \frac{dx^*}{d\underline{K}} < 0 < \frac{dx^*}{dr}, \frac{dx^*}{d\delta}$. Moreover, there is a $\hat{\alpha} \in (0, 1)$ such that $\frac{dx^*}{d\alpha} < 0$ if and only if $\alpha < \hat{\alpha}$.*

The effect of preference for taxation δ , immobile capital $\underline{K}$, and available investment $\bar{K}$ are all through taxes. Higher capital share α has conflicting effects: the direct effect raises investment incentives, but the indirect, strategic effect through higher taxes reduces incentives. When capital share is small, the direct effect tends to dominate, but when it is large, the strategic effect tends to dominate.

Inefficiency If there was only one large investor with capital stock $\bar{K}$ and signal x_i , the investor would invest in the country if and only if

$$\mathbb{E}[\pi(\bar{K}, A) \mid x_i] \geq 0, \quad \text{that is } x_i \geq A_l. \quad (7)$$

However, when there are many investors, strategic uncertainty prevents investors from coordinating their investment decisions efficiently. The following result is immediate from the

¹²It follows that focusing on monotone strategies is without loss of generality and that the unique equilibrium survives the iterated deletion of dominated strategies.

inspection of x^* and A_l .

Proposition 3 (Strategic Uncertainty and Inefficiency). *The investment threshold is lower in the centralized setting with one large investor than in the decentralized setting: $A_l < x^*$. Moreover, $x^* - A_l$ is increasing in α and δ , and decreasing in $\underline{K}$.*

Intuitively, higher capital share α , preference for taxation δ , and lower immobile capital $\underline{K}$ all increase the marginal effect of higher capital investment in reducing tax, e.g., from [Lemma 1](#), we have $\frac{\partial}{\partial \underline{K}} \left| \frac{\partial T^*}{\partial K} \right| < 0$. Similarly, higher available capital $\overline{K}$ implies decisions to invest in the country have higher impact on reducing the effective tax rate.

Alternatively, we note that it is socially optimal for all potential investors to invest when $A \geq A_l$. In contrast, in the decentralized setting, all investors invest if and only if $A \geq x^* + \sigma$. Thus, there is an *inefficiency interval*, $A_I := [A_l, x^* + \sigma)$, such that when productivity is realized within it, $A \in A_I$, there is inefficiently low investment in the country due to strategic uncertainty that hinders coordination among investors.¹³

4 Guaranteed Return with Profit-Sharing (GPS)

[Proposition 3](#) suggests that there is room for intervention by the international community to mitigate the inefficiencies due to politically-induced coordination failure. Subsidy programs can increase investment. For example, from [\(3\)](#), a return subsidy $s > |\alpha \pi(0, \underline{A})|$ will make investment the dominant strategy. However, this induces investment even when investment is inefficient and it is prohibitively expensive because *all* potential investors will always invest to take advantage of government subsidies.

We aim to design subsidy programs that eliminate miscoordination inefficiency in the least costly manner. Such cost-effective programs should, at a minimum, discourage participation by investors who would have invested even without the subsidies. This suggests that optimal

¹³If one large potential investor with signal x_i decides whether to invest, its expected payoff will be $\max \left\{ r, \alpha \mathbb{E}[A|x_i] - \frac{\alpha \delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \overline{K}} \right\} \cdot \overline{K}$, so there is no risk-aversion and no demand for insurance. The only underlying reason for intervention in our setting would be miscoordination inefficiency.

subsidy structure must impose some expected participation costs in addition to providing guaranteed returns.

We begin by considering Guaranteed Return with Profit-Sharing (GPS) programs. GPS programs are described by a guaranteed return $s \in [0, \bar{s}]$, and a profit-sharing rate $t \in [0, 1]$ that a participating investor must pay to the guarantor if the investment turns an economic profit. This is equivalent to programs in which a minimum return is guaranteed and investor pays a premium that is increasing in the profit and is paid only if there is a positive profit. Participants pay the guarantors when profits are sufficiently high, in exchange for receiving payments when losses occur.

We begin with GPS programs for three reasons. First, they take a simple form and may therefore be easier to implement. Second, compared with the GPLS program introduced in the next section, GPS programs require less monitoring of participants' returns. Third, their relative simplicity helps illustrate the key element of screening before turning to the more complex screening mechanisms in GPLS programs.

When GPS programs are offered, the investors' decisions include whether to participate in the program. Accordingly, investor i 's choice set is $k_i \in \{0, 1, 2\}$, where $k_i = 0$ denotes not investing, $k_i = 1$ denotes investing with participation in the program, and $k_i = 2$ denotes investing without participation.

Consider an $(\hat{s}, t)$ program, where $\hat{s}$ is the guarantee and t is the profit-sharing rate. Given productivity A and aggregate investment K , a potential investor i 's returns from not investing, and from investing with and without participation in the program are r , $(1 - T^*)\alpha A + \hat{s} - t \max\{(1 - T^*)\alpha A - r, 0\}$, and $(1 - T^*)\alpha A$, respectively. It is convenient to work with an affine transformation by subtracting the payoffs by r and dividing by α , so that the program becomes $(s, t) = (\hat{s}/\alpha, t)$. Thus, (an affine transformation of) the investor i 's returns from participating in a (s, t) program is

$$\pi(1, K, A) = \pi(K, A) - t \cdot \max\{\pi(K, A), 0\} + s, \quad (8)$$

where $\pi(K, A)$ is defined in (3). The corresponding payoffs from not investing is $\pi(0, K, A) = 0$, and from investing without participation in the program is $\pi(2, K, A) = \pi(K, A)$.

Consistent with the previous section, we assume that an investor chooses to invest in the country when indifferent between investing and not, and chooses not to participate in the program when indifferent between investing with or without participation.

The program makes potential investors more optimistic about their returns from investing in the developing country. Therefore, the lower dominance region (where the dominant strategy is investing in global markets) shrinks. The following assumption ensures the existence of a lower dominance region and the uniqueness of equilibrium with the GPS program. Substantively, it captures the realistic feature that the guarantor does not have so much resources to induce investment even in the worst state of the world $A = \underline{A}$.

Assumption 2. *If productivity is sufficiently low, not investing in the country is the dominant strategy under GPS programs: $\pi(\bar{K}, \underline{A}) + \bar{s} = \underline{A} - A_l + \bar{s} < -\sigma$.*

The condition $\pi(\bar{K}, \underline{A}) + \bar{s} < 0$ suffices for the existence of a lower dominance region. The stricter condition in Assumption 2 ensures a unique equilibrium for a given $\sigma > 0$.

Equilibrium An investor i 's strategy $\sigma_i : [\underline{A} - \sigma, \bar{A} + \sigma] \rightarrow \{0, 1, 2\}$ is a mapping from her signal x_i to her investment and program participation decisions. As in the benchmark, we focus on equilibria in symmetric monotone strategies. A monotone strategy is (weakly) increasing, and hence it is characterized by two cutoffs: there are two thresholds $x' \leq x''$ such that

$$\sigma_i(x_i) = \mathbf{1}_{\{x' \leq x_i < x''\}} + 2 \mathbf{1}_{\{x'' \leq x_i\}},$$

where $\mathbf{1}_{\{\cdot\}}$ is the indicator function. That is, investor i with signal x_i does not invest ($k_i = 0$) in the developing country if $x_i < x'$, invests and participates in the program ($k_i = 1$) if $x_i \in [x', x'')$, and invests without program participation ($k_i = 2$), if $x_i \geq x''$. Without loss of generality $x', x'' \in [\underline{A} - \sigma, \bar{A} + \sigma]$. If it is never (i.e., except on a measure-0 set) optimal for investors to take action 0, we set $x' = \underline{A} - \sigma$. If it is never optimal for investors to take

action 2, we set $x'' = \bar{A} + \sigma$. If investors never take action 1, then $x' = x''$.

Given aggregate investment K and productivity A , investor i 's relative payoffs from a higher versus lower action are

$$\begin{aligned}\delta^1(K, A) &= \pi(1, K, A) - \pi(0, K, A) = \pi(K, A) - t \cdot \max\{\pi(K, A), 0\} + s \\ \delta^2(K, A) &= \pi(2, K, A) - \pi(1, K, A) = t \cdot \max\{\pi(K, A), 0\} - s.\end{aligned}\tag{9}$$

Action and state monotonicity hold. In particular, actions are strategic complements. Given A , both $\delta^1(K, A)$ and $\delta^2(K, A)$ are increasing in K . Similarly, given K , they are increasing in A . Moreover, signals x_i s and the fundamental A are affiliated. Therefore, the game is standard multi-action global game (Frankel, Morris, and Pauzner, 2003; Morris and Shadmehr, 2023, 2024; Van Zandt and Vives, 2007). Proposition 4 characterizes the equilibrium.¹⁴

Proposition 4. *Suppose assumptions 1 and 2 hold and a GPS program (s, t) is in place. As the noise becomes vanishingly small, there is a unique equilibrium. In equilibrium, $x' = A'(s, t) \leq x^*$, where*

$$A'(s, t) = \begin{cases} x^* - s & ; A_l \geq x^* - s \\ \min\{\tilde{A}(s, t), x^*\} & ; A_l \leq x^* - s, \end{cases}$$

and $\tilde{A}$ is the unique solution to

$$\tilde{A} = x^* - s + t \frac{\delta}{(1 - \alpha)^2 \bar{K}} \left\{ \frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} - \log \left(\frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} \right) - 1 \right\}.$$

Moreover,

- if $s \geq t(\bar{A} - A_l)$, then $x'' = \bar{A}$, so that all who invest will participate in the program.
- if $s < t(\bar{A} - A_l)$, then $x'' = A''(s, t) < \bar{A}$, where

$$A''(s, t) = \begin{cases} A_l + s/t & ; s \geq t(A'(s, t) - A_l) \\ A'(s, t) & ; s \leq t(A'(s, t) - A_l), \end{cases}$$

¹⁴We characterize the equilibrium with finite noise in Proposition 10 in Online Appendix A.

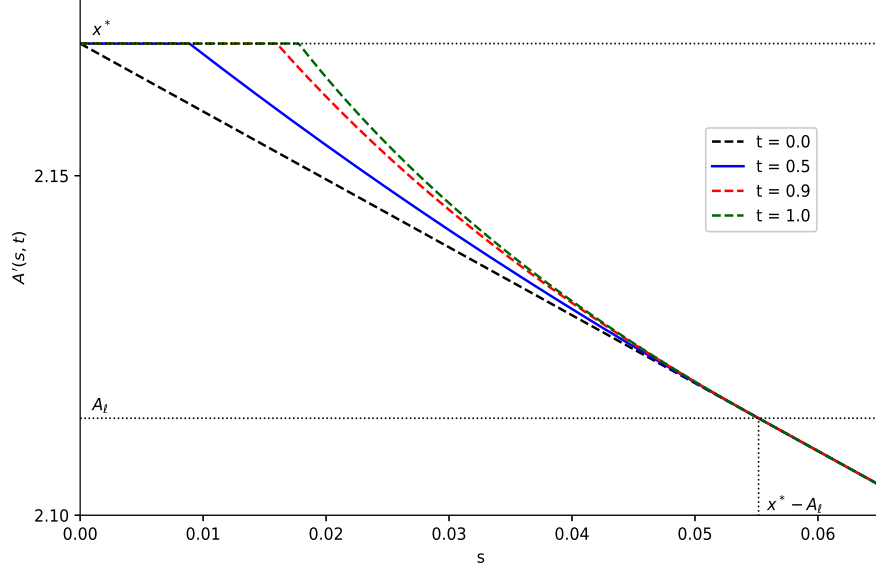


Figure 1: The equilibrium investment threshold $A'(s, t)$ as a function of the GPS program's guarantee s and profit-sharing rate t . Note that $A'(s = x^* - A_l, t) = A_l$, where A_l and x^* are defined in [Assumption 1](#) and [Proposition 2](#), respectively. Parameters: $\delta = 0.1$, $\bar{s} = 0.15$, $\alpha = 0.5$, $r = 1$, $\underline{K} = \underline{A} = 1.5$, $\bar{K} = 2$, $\bar{A} = 2.27$.

so that some who invest will not participate in the program.

When $s = t = 0$, [Proposition 4](#) implies $A' = x^*$ and $A'' = \bar{A}$. The choice of A'' is inconsequential because there is no difference between participating in such a program or not. Thus, there is indeterminacy in A'' , and we have picked a convenient A'' for this special case. For example, if $s, t \rightarrow 0$, with $s/t < \bar{A} - A_l$, then A'' approaches a threshold strictly less than $\bar{A}$.

[Proposition 4](#) has intuitive features. The threshold for investing in the country, A' , is weakly lower than the threshold absent any program; strictly so when the guarantee s is sufficiently large. [Figure 1](#) demonstrates—in a setting where $\bar{s} > x^* - A_l$. When the guarantee s is sufficiently small ($s \approx 0 < x^* - A_l$), we have $A' = x^*$ regardless of the profit-sharing magnitude t , and the program has no effect, because no one participates. As the guarantee s increases, A' falls below x^* along $\tilde{A}(s, t)$, reaching $A' = A_l$ (at $s = x^* - A_l$), the productivity threshold below which investors have a dominant strategy not to invest absent programs. At this threshold, the coordination failure is fully resolved. Once the guarantee s exceeds the

threshold $x^* - A_l$, the investment threshold becomes $A' = x^* - s < A_l$, falling with s .

The logic is that the marginal investor with threshold signal believes that the fraction of investors who will invest is uniformly distributed on $[0, 1]$. From (6), the anticipated (normalized) tax payment (per unit of capital) associated with that belief is $x^* - r/\alpha$. When $t = 0$, as the guarantee s increases, it compensates for both this tax and for a lower productivity, thereby pushing down the investment threshold. In particular, from (8),

$$\begin{aligned} \mathbb{E}_A[\pi(1, K(A; A'), A) \mid x_i = A'] &\approx A' - \frac{\mathbb{E}_K[\text{tax payment}]}{\alpha} \Big|_{K \sim U[0, \bar{K}]} - \frac{r}{\alpha} + s = A' - x^* + s \\ &= A' - x^* + s, \end{aligned} \quad (10)$$

where tax payment is per unit of capital. At $s = 0$, the investment threshold at which the investor is indifferent between investing in the country and not is x^* . As s increases, this threshold falls linearly with s .

Figure 1 also shows how the profit-sharing rate t affects the investment threshold A' . When there is profit-sharing $t > 0$, the threshold tends to be higher for a given guarantee, because some of an investor's profit will be clawed back. However, the profit-sharing magnitude t is irrelevant when the guarantee is so small that no one participates, or so large that the marginal investor anticipates to make no profit to share, investing only to collect the guarantee. In the middle, where $A' \in (A_l, x^*)$, higher profit-sharing naturally increases the investment threshold A' . We will describe the intuition for A'' in the context of optimal GPS following Proposition 5.

To reach $A' = A_l$, the program needs a sufficiently high guarantee $s = x^* - A_l$. However, Assumption 2 places an upper bound on the guarantee, preventing the guarantor from inducing investment in the worst state of the world: when the noise is vanishingly small, it requires that $s \leq \bar{s} < A_l - \underline{A}$. Assumption 3 ensures that, for a fixed noise level σ , it is feasible to resolve the coordination failure, without violating Assumption 2: the guarantor cannot induce investment in the worst state of the world, but it can fully remove coordination failure.¹⁵

¹⁵Given σ , there is no coordination failure whenever $x' \leq A_l - \sigma$.

Assumption 3. *It is feasible to remove the inefficiency interval: $\sigma < A_l - (\underline{A} + x^*)/2$*

Because $x^* > A_l$ by [Proposition 3](#), the bound $A_l - (\underline{A} + x^*)/2 > 0$ for sufficiently small $\underline{A}$, so that [Assumption 3](#) would hold when noise is small.¹⁶

To eliminate coordination failure, the guarantor must induce potential investors to invest whenever their signals are above the threshold A_l —when the noise is vanishingly small. [Proposition 4](#) shows that this can be achieved by providing a guarantee $s = x^* - A_l$, regardless of the profit-sharing magnitude—see [Figure 1](#). It follows that the cheapest program that can eliminate miscoordination must set the guarantee $s = x^* - A_l$ and the profit-sharing magnitude $t = 1$, both to maximize the clawback and to disincentivize program participation by those who invest in the country even without additional incentives. [Proposition 5](#) formalizes this result and computes the associated cost of the optimal GPS program that eliminates miscoordination.

Proposition 5. *Suppose assumptions [1](#), [2](#), and [3](#) hold. As the noise becomes vanishingly small, the optimal GPS that eliminates miscoordination features $(s, t) = (x^* - A_l, 1)$, and its expected cost (per unit of capital) is $\frac{1}{2} \frac{(x^* - A_l)^2}{A - \underline{A}}$.*

The intuition builds on that of [Proposition 4](#) around Equation (10). Absent programs, the opportunity cost of investing for the marginal investor with the threshold signal x^* includes the normalized tax payment (per unit of capital) associated with the belief that the fraction of investors who will invest is uniformly distributed on $[0, 1]$:

$$\left. \frac{\mathbb{E}_K[\text{tax payment}]}{\alpha} \right|_{K \sim U[0, \bar{K}]} = x^* - \frac{r}{\alpha}. \quad (11)$$

The guarantor wants the marginal investor to behave *as if* she believes that all other potential

¹⁶[Assumption 3](#) tends to require that $\underline{A}$ be small, while [Assumption 1.1](#) tends to require that it be large—to ensure interior tax rate. These assumptions can be satisfied simultaneously when, e.g., r and $\bar{A}$ are sufficiently large that $\frac{r}{\alpha} > \frac{\delta}{(1-\alpha)^2} \left\{ \frac{1}{\bar{K} + \underline{K}} \frac{\bar{K} - \underline{K}}{\underline{K}} + \frac{1}{\bar{K}} \log \left(\frac{\bar{K} + \bar{K}}{\underline{K}} \right) \right\} + 2\sigma$ and [Assumption 1.3](#) hold, respectively.

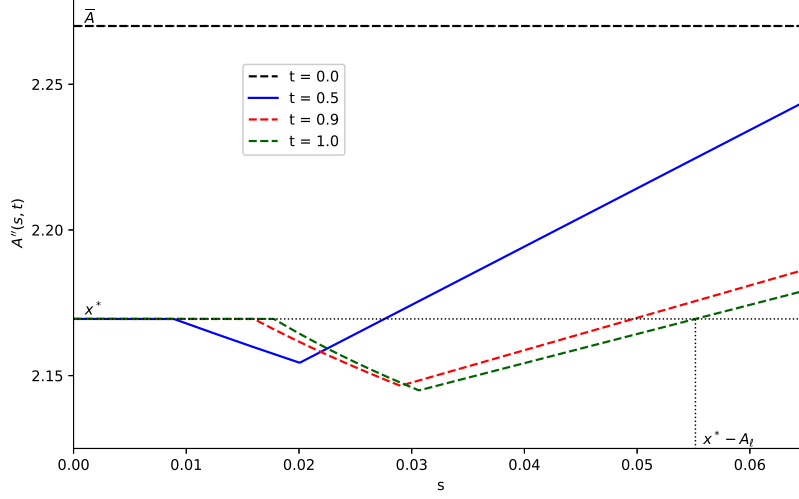


Figure 2: The equilibrium participation threshold $A''(s, t)$. Parameters: $\delta = 0.1$, $\bar{s} = 0.15$, $\alpha = 0.5$, $r = 1$, $\underline{K} = \underline{A} = 1.5$, $\bar{K} = 2$, $\bar{A} = 2.27$.

investors will invest, so that the associated tax payment is only

$$\frac{\mathbb{E}_K[\text{tax payment}]}{\alpha} \Big|_{K=\bar{K}} = A_l - \frac{r}{\alpha}. \quad (12)$$

The guarantor pays the difference, $x^* - A_l$, via the guarantee. Moreover, because there is no (pre-program) profit at $A = A_l$, profit-sharing rate is irrelevant for the marginal investor:

$$\mathbb{E}_A[\pi(K(A; A_l), A)|x_i = A_l] < \mathbb{E}_A[\pi(\bar{K}, A)|x_i = A_l] = 0.$$

While the magnitude of profit-sharing rate t does not change the marginal investor's decision whether to invest in the country, it does change program participation decisions of those who invest. Therefore, it plays a critical role in program costs and hence the design of optimal GPS programs. This is apparent in [Figure 2](#): among GPS programs with guarantee $s = x^* - A_l$, which resolve miscoordination, (non-)participation threshold A'' varies with the profit-sharing rate t . A higher threshold A'' means that a larger fraction of investors participate in the program. In particular, when $t < 1$, so that $A'' > x^*$, investors with signals $x_i \in (x^*, A'')$ participate, even though they would have invested in the country even absent

any program. The participation of these investors does not help with miscoordination, but raises the costs of the program. Profit-sharing is crucial to screen them out.

Remarkably, by setting $t = 1$ and thereby claiming all pre-program profits, a GPS program can nearly perfectly screen out investors who would have invested even without the program—inducing the participation threshold $A'' = x^*$. Intuitively, absent programs, the marginal investor with signal x^* is indifferent between investing in the country or in global markets. Participation in the program $(s, t) = (x^* - A_l, 1)$, provides this marginal investor (with signal x^*) the payoff $s = x^* - A_l$, regardless of any pre-program profits which will be fully taken away at the profit-sharing rate of 1. By contrast, not participating in the program simply makes this marginal investor more optimistic about aggregate investment and hence the anticipated tax payment: the marginal investor with signal x^* believes that almost all investors will invest, because $x^* > A_l$, the new investment threshold induced by the program. The difference in the anticipated tax payment is exactly $x^* - A_l$ as we discussed—see (11) and (12). That is, with program $(s, t) = (x^* - A_l, 1)$, the marginal investor with the threshold signal x^* is indifferent between investing with and without participation in the program. It follows that reducing the profit-sharing rate from $t = 1$ tips the balance in favor of participation, raising the participation threshold A'' above x^* , and hence program costs.

These intuitions also allow us to provide a heuristic derivation of the program costs. Because investors with signals $x_i \in [A', A'']$ participate in the program, when the noise is vanishingly small, almost all investors participate in the program with probability $\frac{A'' - A'}{A - \underline{A}} = \frac{x^* - A_l}{A - \underline{A}}$. Moreover, an investor i with signal $x_i \in (A', A'')$ obtains a net transfer

$$s - t \pi(\bar{K}, A) \approx (x^* - A_l) - 1 (x_i - A_l) = x^* - x_i = A'' - x_i > 0, \quad (13)$$

so that $\mathbb{E}[A'' - x_i \mid x_i \in (A', A'')] \approx \mathbb{E}[x^* - A \mid A \in (A_l, x^*)] = x^* - \frac{A_l + x^*}{2} = \frac{x^* - A_l}{2}$. Therefore, the expected costs are

$$\Pr(x_i \in (A', A'')) \cdot \mathbb{E}[A'' - x_i \mid x_i \in (A', A'')] \approx \frac{1}{2} \frac{(x^* - A_l)^2}{A - \underline{A}},$$

as specified in [Proposition 5](#).

[Corollary 2](#) is immediate from [Proposition 3](#) and [Proposition 5](#).

Corollary 2. *The expected cost of the optimal GPS that eliminates miscoordination is increasing in preference for taxation δ and capital share α , and it is decreasing in the immobile capital level $\underline{K}$.*

This result suggests that eliminating miscoordination via GPS programs is cheaper when capital is invested in industries with less advanced automation technology ([Acemoglu and Restrepo, 2018](#)) (lower α), or where there is abundant immobile capital (higher $\underline{K}$).

Proposition 6. *The term $(x^*(\bar{K}) - A_l(\bar{K}))^2$, and hence the expected cost (per unit of capital) of the optimal GPS, is strictly single-peaked in available capital stock for investment $\bar{K}$. Moreover, for sufficiently large $\bar{K}$, the total cost of the program is decreasing in $\bar{K}$.*

[Proposition 6](#) shows that the optimal GPS exhibits economies of scale. In particular, subsidy costs per unit of capital are single-peaked in the level of capital investment, reaching their maximum at an intermediate level. The implication is that development agencies may benefit from setting sufficiently ambitious investment targets.¹⁷

To see the intuition, note that the subsidy per unit of capital is proportional to $x^*(\bar{K}) - A_l(\bar{K})$, the gap between the normalized tax rate absent intervention, $T_1(\bar{K}) := x^*(\bar{K}) - r/\alpha$, and the normalized tax rate if the entire available capital stock is invested, $T_2(\bar{K}) := A_l(\bar{K}) - r/\alpha$. A sufficient condition for $T_1(\bar{K}) - T_2(\bar{K})$ to be single-peaked is that the ratio of marginal effects, $T_1'(\bar{K})/T_2'(\bar{K})$, be increasing.¹⁸ An increase in capital stock $\bar{K}$ generates two competing forces. First, strategic uncertainty lowers T_2 faster than T_1 : absent intervention, only a fraction of available capital is invested. Second, the marginal effect of investment on political risk diminishes: as taxes fall, further reductions become increasingly difficult. When capital

¹⁷As [Lemma 5](#) in the Online Appendix shows, for $(x^* - A_l)^2$ to increase for small $\bar{K}$, it suffices that the tax rate decline with $\bar{K}$; for it to decrease for large $\bar{K}$, it suffices that the marginal effect of capital on the tax rate converges to zero faster than $1/x$. Single-peakedness requires that the solution to $\int_0^k (g(x) - g(k)) dx/k = -kg'(k)$ be unique, where $g(\cdot)$ maps $\bar{K}$ to the tax rate, e.g., $g(K) = 1/(\underline{K} + K)$.

¹⁸Because $T_1'(\bar{K}) - T_2'(\bar{K}) = T_2'(\bar{K})(T_1'(\bar{K})/T_2'(\bar{K}) - 1)$ and both T_1 and T_2 are decreasing in $\bar{K}$.

stock is low and T_1 and T_2 remain relatively close, the first force dominates. As capital stock grows and T_2 becomes substantially smaller than T_1 , the second effect dominates. The last part follows because the costs per unit of capital fall at a rate faster than $1/x$.

5 Guaranteed Return with Profit- and Loss-Sharing (GPLS)

The optimal GPS program has appealing features. First, it takes a simple form: a minimum return is guaranteed, and returns above a maximum are clawed back. The guarantor effectively rents capital from investors who are willing to participate at a rate exceeding the global market return and then invests it in the developing country. Second, the program screens out all investors who would have invested even in the absence of the program.

However, as equation (13) showed, even the optimal GPS program offers net positive subsidy to all participants except the most *optimistic* one with signal $x_i = A''$, who is made indifferent between investing with and without participation. In fact, all participants will end up with the same, strictly positive net payoff $s = x^* - A_l$ from investing in the country versus global markets. This feature is problematic for two reasons. First, ideally a GPS program should make participants indifferent between investing and not investing in the country, instead of leaving them with net positive payoffs that the program must then pay. Second, taking a mechanism design approach, more optimistic investors have more incentives to misreport their types (signals) and therefore should receive (weakly) more rent (Börger, 2015, Ch. 1). Given that the optimal GPS program already required profit-sharing at rate 1, any adjustment must reduce the guarantee. However, the guarantee is already at the minimum amount required to eliminate miscoordination inefficiencies. This suggests that the GPS program's instruments are too coarse. To reduce costs, a program should either (1) allow some miscoordination inefficiencies, or (2) introduce new instruments that can provide different compensation to different investors depending on their types (signals) or equivalently, their expected pre-program payoffs.

We thus extend the GPS program to Guaranteed Return with Profit- and Loss-Sharing

(GPLS) program by introducing loss-sharing at a rate t^- . GPLS programs are described by a guaranteed return $s \in [0, \bar{s}]$, a profit-sharing rate $t^+ \in [0, 1]$, and a loss-sharing rate $t^- \in [0, \bar{t}]$ with $\bar{t} < 1$, which specifies the fraction of pre-program losses covered by the guarantor. The upper bound $\bar{t} < 1$ captures realistic settings in which the guarantor is unable to induce investment in the worst state of the world.

Payoffs from not investing and from investing without participation in the program as well as our tie-breaking rules are the same as before. Likewise, [Assumption 2](#) is modified to

Assumption 4. *If productivity is sufficiently low, not investing in the country is the dominant strategy under GPLS program: $\pi(\bar{K}, \underline{A}) + \frac{\bar{s}}{1-\bar{t}} = \underline{A} - A_l + \frac{\bar{s}}{1-\bar{t}} < -\sigma$.*

Because GPLS programs offer loss-sharing in addition to the guarantee, [Assumption 4](#) implies [Assumption 2](#), by setting $\bar{t} = 0$.

Analogously to (9), given productivity A and aggregate investment K , the incremental payoffs from taking higher versus lower actions under a GPLS program (s, t^+, t^-) are

$$\begin{aligned}\delta_{gpls}^1(K, A) &= \pi(K, A) - t^+ \max\{\pi(K, A), 0\} - t^- \min\{\pi(K, A), 0\} + s \\ \delta_{gpls}^2(K, A) &= t^+ \max\{\pi(K, A), 0\} + t^- \min\{\pi(K, A), 0\} - s.\end{aligned}\tag{14}$$

Action and state monotonicity continue to hold, and hence best responses to monotone strategies are also monotone. Similarly, we can compute the equilibrium cutoffs x' and x'' under a GPLS program analogously to a GPS program.¹⁹

Proposition 7. *Suppose assumptions 1 and 4 hold and a GPLS program (s, t^+, t^-) is in place. As the noise becomes vanishingly small, there is a unique equilibrium. In equilibrium, $x' = A'(s, t^+, t^-) \leq x^*$, where*

$$A'(s, t^+, t^-) = \begin{cases} x^* - \frac{s}{1-t^-} & ; A_l \geq x^* - \frac{s}{1-t^-} \\ \min\{\tilde{A}(s, t^+, t^-), x^*\} & ; A_l \leq x^* - \frac{s}{1-t^-}, \end{cases}$$

¹⁹We characterize the equilibrium with finite noise in [Proposition 11](#) in Online Appendix A.

and $\tilde{A}$ is the unique solution to

$$\tilde{A} = x^* - \frac{s}{1-t^-} + \frac{t^+ - t^-}{1-t^-} \frac{\delta}{(1-\alpha)^2 \bar{K}} \left\{ \frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} - \log \left(\frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} \right) - 1 \right\}.$$

Moreover,

- if $s \geq t^+ (\bar{A} - A_l)$, then $x'' = A''(s, t^+, t^-) = \bar{A}$, so that all who invest will participate in the program.
- if $s < t^+ (\bar{A} - A_l)$, then $x'' = A''(s, t^+, t^-) < \bar{A}$, where

$$A''(s, t^+, t^-) = \begin{cases} A_l + s/t^+ & ; s \geq t^+ (A'(s, t^+, t^-) - A_l) \\ A'(s, t^+, t^-) & ; s \leq t^+ (A'(s, t^+, t^-) - A_l), \end{cases}$$

so that some who invest will not participate in the program.

Proposition 7 becomes Proposition 4 by setting $t^- = 0$ and $t^+ = t$. From Proposition 7, $A'' = \bar{A}$ whenever $s = t^+ = 0$ regardless of t^- : When $t^- > 0$, loss-sharing benefits the participants. When $t^- = 0$, participation has no effect. Thus, there is indeterminacy in A'' , and we have picked a convenient A'' for this special case as we did for GPS programs.

To reach $A' = A_l$, GPLS programs incentivize investors by a sufficiently attractive combination of guarantee and lost-sharing: $s/(1-t^-) = x^* - A_l$.²⁰ The logic is similar to our discussion following Proposition 5. The guarantor must pay $x^* - A_l$ to the marginal investor, so that she invests when her signal is just above A_l . With GPLS programs, the guarantor can do so through two channels: guarantee and loss-sharing. With a loss-sharing rate of t^- , the guarantor transfers $t^- (x^* - A_l)$ to the marginal investor through loss-sharing, requiring a guarantee of $s = (1-t^-)(x^* - A_l)$ to cover the rest.

The key insight of the GPLS program is that loss sharing is an effective screening tool, whereas guarantees are not. Any combination of a guarantee s and a loss-sharing rate t^-

²⁰ Assumption 4 places an upper bound on the guarantee. Assumption 3 ensures that $x^* - A_l < A_l - \underline{A}$, so that miscoordination can be eliminated even when the guarantor cannot induce coordination is the worst state of the world.

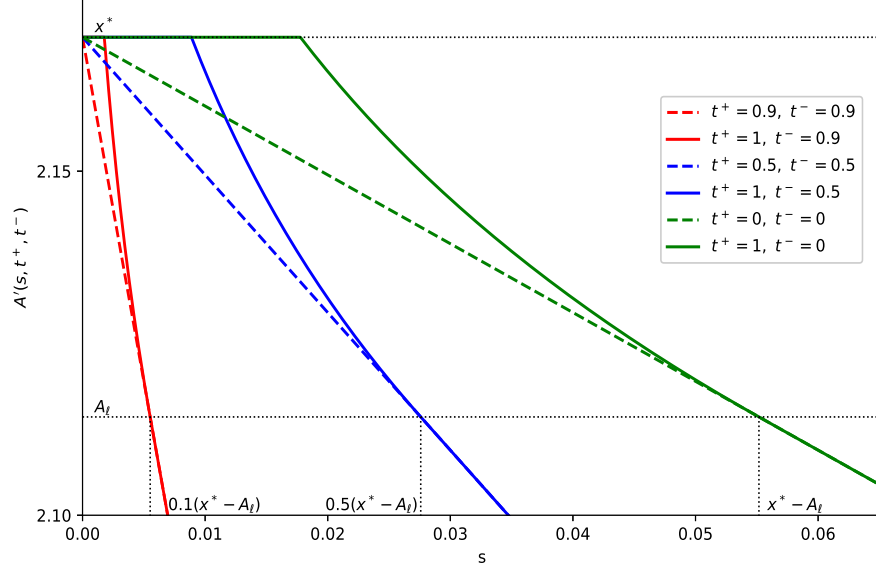


Figure 3: The equilibrium investment threshold $A'(s, t^+, t^-)$ as a function of the GPLS program's guarantee s , profit-sharing parameter t^+ , and loss-sharing t^- . Parameters: $\delta = 0.1$, $\bar{s} = 0.15$, $\alpha = 0.5$, $r = 1$, $\underline{K} = \underline{A} = 1.5$, $\bar{K} = 2$, $\bar{A} = 2.27$, $\bar{t} = 0.9$.

satisfying $s/(1 - t^-) = x^* - A_l$ eliminates miscoordination, as it makes no difference to an investor whether compensation comes through the guarantee or through loss sharing. As illustrated in Figure 3, $A' = A_l$ when $s = (1 - t^-)(x^* - A_l)$ for $t^- = 0, 0.5$, or 0.9 . To screen out more optimistic investors and minimize costs, the guarantor should maximize the loss-sharing rate and set the guarantee just high enough to induce the marginal investor to invest. More optimistic investors anticipate smaller or no losses—and therefore little or no subsidy—through loss sharing. In contrast, every participant expects to receive the same guarantee. It follows that the optimal GPLS program features $t^- = \bar{t}$ and $s = (1 - \bar{t})(x^* - A_l)$. Mirroring the logic of the GPS program, the guarantor also sets $t^+ = 1$.

Proposition 8. *Suppose assumptions 1, 3, and 4 hold. As the noise becomes vanishingly small, the optimal GPLS program that eliminates miscoordination is $(s, t^+, t^-) = ((1 - \bar{t})(x^* - A_l), 1, \bar{t})$. Its expected cost (per unit of capital) is $\frac{(1 - \bar{t})^2}{2} \frac{(x^* - A_l)^2}{\bar{A} - \underline{A}}$, which is strictly smaller than the expected costs of the optimal GPS program if and only if $\bar{t} > 0$.*

The optimal GPLS program's expected cost is lower than that of the optimal GPS program

and coincides with it for $t^- = 0$. Remarkably, when loss-sharing rate is near 100%, the program costs are negligible—in the limit as the noise vanishes. The intuition is that the guarantor does not need to provide a guarantee to induce the marginal investor to invest: when $t^- = \bar{t} \approx 1$, the optimal guarantee $s = (1 - \bar{t})(x^* - A_l) \approx 0$. With negligible optimal guarantee, and a positive profit-sharing rate $t^+ = 1$, nearly no investor participates in the program: $A'' \approx A' \approx A_l$: investors with signals slightly more optimistic than the marginal investor—those with signals $x_i \in (A_l, A_l + \epsilon)$ —anticipate a positive payoff, $\pi(\bar{K}, A_l + \epsilon) \approx \epsilon > 0$, and hence no loss-sharing subsidy and nearly no guarantee $s \approx 0$, but they have to share their profit as $t^+ = 1$.²¹ The logic builds on complementarities among investors. When the most pessimistic investors participate in the program and invest, they tip the balance for the more optimistic investors who were on the fence, prompting them to invest as well. Their participation, in turn, induces the next group of slightly less optimistic investors to follow, and so on.

The same logic applies in the more realistic setting with $\bar{t} < 1$. The program leverages complementarities among investors to stimulate investment not through large subsidies, but by providing just enough support to create a widespread belief that others will invest.

Finally, combining Propositions 6 and 8 shows that the expected costs per unit of capital are single-peaked in the available capital stock for investment, $\bar{K}$, implying economies of scale in investment programs induced by strategic uncertainty.

5.1 Restricted GPLS (Shen-Zou Program)

GPLS programs allow for different rates for profit- and loss-sharing. We now analyze the additional costs to the guarantor introduced if we restrict the rates to be equal: $t^+ = t^- \leq$

²¹We will discuss realistic restrictions on $\bar{t}$ in the next subsection. Here, we clarify the technicalities concerning $\bar{t} \approx 1$ for completeness. Assumption 4 rules out $\bar{t} = 1$ for substantive reasons: the guarantor should not be able to induce investment even in the worst state of the world. If it was allowed, $\bar{t} = 1$, $s = 0$, and any $t^+ > 0$ would induce a continuum of equilibria with $A'' = A_l$ and $A' \in [\underline{A}, A_l]$ without our tie-breaking rule, and $A' = \underline{A}$ with it. Then, an optimal GPLS program along the following path $\bar{t} \rightarrow 1$, $s = (1 - \bar{t})(x^* - A_l) \rightarrow 0$, and any $t^+ > 0$ can be viewed as selecting the equilibrium with $A' = A'' = A_l$ when first $\sigma \rightarrow 0$ and then $\bar{t} \rightarrow 1$. However, for any given $\bar{t} < 1$, the optimal guarantee $s > 0$, and hence the optimal profit-sharing $t^+ = 1$.

$\bar{t} < 1$. We refer to such programs as Restricted GPLS (R-GPLS) programs. An R-GPLS program is analogous to the intervention that [Shen and Zou \(2024\)](#) suggest in the context of bank runs, and therefore we also refer to them as SZ programs.

Proposition 9. *Suppose assumptions [1](#), [3](#), and [4](#) hold. As the noise becomes vanishingly small, the optimal R-GPLS (SZ) program that eliminates miscoordination is $(s, t^+, t^-) = ((1 - \bar{t})(x^* - A_l), \bar{t}, \bar{t})$. The expected cost for the R-GPLS (SZ) program is strictly larger than the expected costs of the optimal GPLS program if and only if $\bar{t} < 1$.*

Intuitively, for any $\bar{t} < 1$, the optimal program requires $s > 0$, creating incentives for optimistic investors to participate. To screen them out, the guarantor wants to choose the highest profit-sharing rate t^+ , and so it chooses its upper bound $\bar{t} < 1$. In the GPLS program this upper bound is 1—without violating any substantive assumption, e.g., [Assumption 4](#). Therefore, the costs of R-GPLS programs are higher.

6 Comparisons: GPS, GPLS, SZ, and Common Programs

Propositions [5](#), [8](#), and [9](#) allow us to compare the costs associated with the programs that aim to eliminate miscoordination inefficiencies.

Corollary 3. *Suppose Assumptions [1](#) to [4](#) hold. Let $C(i)$ be the expected costs (per unit of capital) of the optimal program $i \in \{GPS, GPLS, SZ\}$. As the noise becomes vanishingly small,*

$$C(GPLS) = m(\bar{t}) \cdot C(SZ) = (1 - \bar{t})^2 \cdot C(GPS),$$

where $m(\bar{t}) \leq \max\{\bar{t}, \frac{x^* - A_l}{A - A_l}\}$. That is, the costs of optimal SZ and GPS programs are larger than that of the optimal GPLS program by factors of $1/m(\bar{t})$ and $1/(1 - \bar{t})^2$, respectively.

When there is nearly complete loss-sharing, GPLS and SZ (R-GPLS) coincide: both $t^- \approx t^+ = 1$. However, this would imply that the guarantor can induce investment regardless of how bad the state of the world will be—recall [Assumption 4](#). A more realistic assumption is that $\bar{t}$ is bounded away from 1. Then, [Corollary 3](#) shows that restricted GPLS programs

return	$A < A_l$	$A > A_l$
r_{gps}	$r - \alpha(A_h - x^*) - \alpha(A_l - A)$	$r + \alpha(x^* - A_l)$
r_{gpls}	$(1 - \bar{t}) r_{\text{gps}} + \bar{t} r$	$r + (1 - \bar{t}) \alpha(x^* - A_l)$
r_{np}	$r - \alpha(A_h - A)$	$r + \alpha(A - A_l)$

Table 1: Investor absolute returns under optimal GPS and GPLS programs.

(SZ programs) by requiring $t^- = t^+$ raises expected program costs by a factor of $1/m(\bar{t})$.²²

Conversely, relaxing the R-GPLS (SZ) programs by allowing full profit-sharing while maintaining the realistic assumption of restricted loss-sharing rate to a maximum of $\bar{t} < 1$ can generate significant savings, more so when $\bar{t}$ is lower or when miscoordination is a more significant problem, so that $x^* - A_l$ is larger; that is when preference for taxation δ or capital share α are higher, or the immobile capital level $\underline{K}$ is lower—see [Proposition 3](#).

We performed the analysis in normalized and net returns, because they determine strategic behavior. We end by showing the investors' absolute returns in equilibrium (when the noise is vanishingly small) under the optimal GPS and GPLS programs.

Corollary 4. *Let r_{gps} and r_{gpls} denote an investor's (absolute) return from participating in the optimal GPS and GPLS programs, respectively, and let r_{np} denote the corresponding return for a non-participating investor. Then, returns are given in [Table 1](#).*

Figure 4 illustrates. Optimal GPS and GPLS programs both feature a minimum guaranteed return. Importantly, they both also feature a *maximum* return. Participating investors get a higher payoff than non-participating ones if and only if the productivity shock A is sufficiently low. For the optimal GPLS program this threshold is higher than A_l only by a margin of $s_{\text{gpls}} = (1 - \bar{t})s_{\text{gps}}$, where s_{gpls} and s_{gps} are the guarantees under the optimal GPLS and GPS programs, respectively. Therefore, when $\bar{t}$ is only slightly below 1, potential investors who believe that productivity is at least slightly above A_l will not participate, reducing the

²²SZ (R-GPLS) programs are more cost effective than GPS programs if and only if $\bar{t}$ is above a threshold. The threshold is the solution to $(1 - x)^2 = m(x)$ in the interval $(0, 1)$.

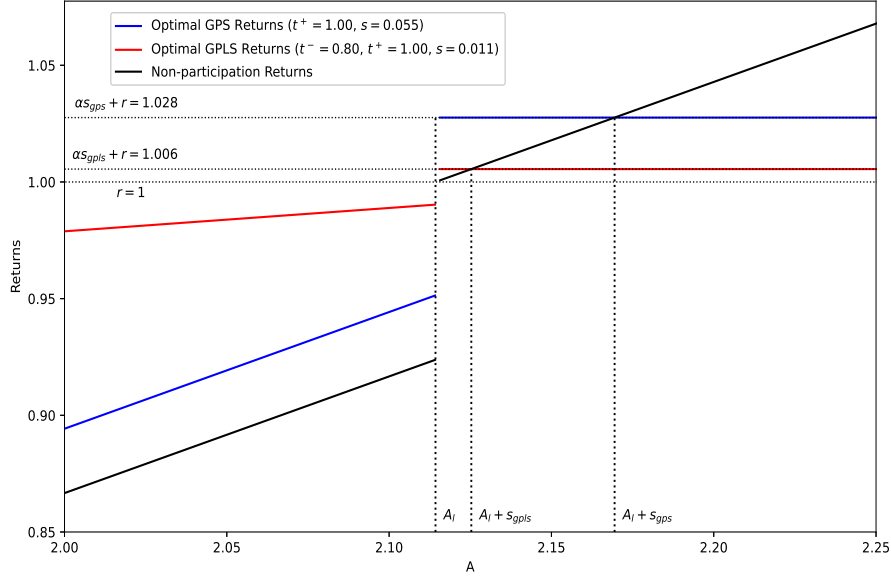


Figure 4: Absolute returns as a function of productivity A under optimal GPS and GPLS programs with guarantees s_{gps} and s_{gpls} , respectively. Parameters: $\delta = 0.1$, $\bar{s} = 0.15$, $\alpha = 0.5$, $r = 1$, $\underline{K} = \underline{A} = 1.5$, $\bar{K} = 2$, $\bar{A} = 2.27$, $\bar{t} = 0.8$, $A' = A_l$.

program costs. Critically, while the programs look qualitatively similar, as [Corollary 3](#) shows, even at a loss-sharing rate of $t^- = 0.5$, the optimal GPLS program costs only a quarter of the optimal GPS program. The difference is in better screening: the optimal GPLS program will have far less participants than the optimal GPS program, underlying its efficiency.

6.1 Common Programs: Political Risk Insurance, Concessional Loans, and Blended Finance

International organizations, national governments, and private-sector organizations, provide instruments to mitigate political risk. While the international community is aware of the coordination problem among investors (the academic literature goes back to at least [Rosenstein-Rodan \(1943\)](#)) and the reports of development agencies engage with the issue ([World Bank, 2017](#)), available instruments typically focus on addressing political risk for one or few investors and creditors, overlooking the coordination problem. We present an overview of these instruments and show how their structures relate to the optimal GPLS

program structure. This comparison provides a benchmark that highlights the features of the optimal GPLS program, and brings to focus how existing programs could be adjusted to address miscoordination inefficiencies.

Two classes of instruments stand out: political risk insurance and concessional loans.²³

Political Risk Insurance compensates for losses caused by covered adverse political events (e.g., breach of contract or political violence) at a premium. These products are not subsidies in general, and the agencies that offer them are self-sustaining. Examples include (Multilateral Investment Guarantee Agency) MIGA’s *Guarantees* and (U.S. International Development Finance Corporation) DFC’s *Political Risk Insurance*.²⁴ ([Multilateral Investment Guarantee Agency, 2024](#); [World Bank, 2016](#)).

Concessional Loans to investors below the market rate. For example, (International Finance Corporation) IFC’s blended finance instrument offers concessional loans to investors through the International Development Association (IDA) Private Sector Window (PSW).²⁵

Instruments may be combined in what is known as *blended finance*, in which, for example, the International Finance Corporation provides loans, guarantees, and equity investments—some or all of which may be offered at a discount.²⁶

A concessional loan reduces an investor’s cost of capital and resembles the guaranteed return in our setting, which is $\alpha s_{gpls} = \alpha(1 - t^-)(x^* - A_l)$ for the optimal GPLS program. Political risk insurance compensations or guarantees are similar to loss-sharing. In [Table 1](#), we can

²³Other policies include providing technical assistance, making equity investment, and offering grants, although major agencies tend to offer grants only to governments.

²⁴See <https://www.dfc.gov/what-we-offer/our-products> for DFC and <https://www.miga.org/products> for MIGA.

²⁵See <https://ida.worldbank.org/en/financing/ida-private-sector-window> for PSW and <https://www.ifc.org/en/what-we-do/sector-expertise/blended-finance/how-blended-finance-works> for IFC’s Blended Finance.

²⁶These discounts are sometimes referred to as concessionality—see <https://www.ifc.org/en/what-we-do/sector-expertise/blended-finance/how-blended-finance-works#concessionality>. For an example, see *Ghana Sankofa Gas Project* (2020), in which the World Bank Group used several instruments to help mobilize about \$8 billion in private investments for an offshore natural gas development project in Ghana.

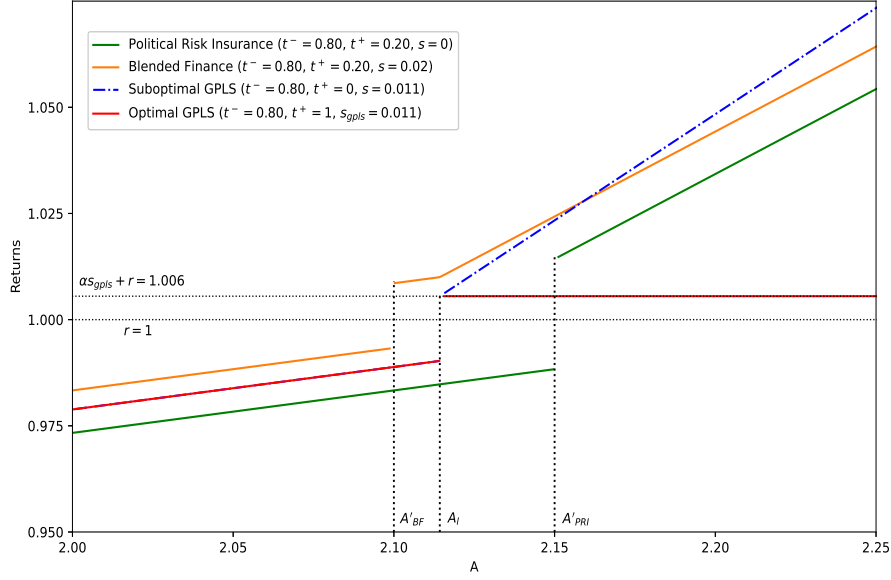


Figure 5: Absolute returns as a function of productivity A for Political Risk Insurance (PRI), Blended Finance (BF), a suboptimal GPLS, and the optimal GPLS. A'_{BF} and A'_{PRI} denote the investment thresholds (A 's) under BF and PRI, respectively. Parameters: $\delta = 0.1$, $\bar{s} = 0.15$, $\alpha = 0.5$, $r = 1$, $\underline{K} = \underline{A} = 1.5$, $\bar{K} = 2$, $\bar{A} = 2.27$, $\bar{t} = 0.8$.

write r_{gpls} when $A < A_l$ (so there is a loss) as $r + \alpha s_{gpls} - (1 - t^-)\alpha(A_h - A)$. Then $\alpha(A_h - A)$ may be viewed as the loss and $\bar{t}$ as the loss-sharing rate.²⁷ Similarly, when $A > A_l$, so there is a profit, r_{gpls} can be written as $r_{np} + \alpha s_{gpls} - \alpha(A - A_l)$, so that $\alpha(A - A_l)$ resembles the premium paid to the agency. However, “premium” here is state-contingent, not a fixed, predetermined amount.

In sum, one may think of political risk insurance and “guarantee programs”, in a stylized sense, as featuring $t^-, t^+ \in (0, 1)$ and $s = 0$, and concessional loans as $s > 0$ with $t^- = t^+ = 0$.

Figure 5 shows how a participating investor’s absolute return under the optimal GPLS program—designed to eliminate miscoordination—compares with alternatives that are suboptimal for addressing coordination risk. For example, political risk insurance without concessional loans ($s = 0$) cannot eliminate miscoordination. Programs with $t^+ < 1$ are significantly more expensive: not only they offer higher payments to each participant, but importantly,

²⁷If the program accounts for “concessional” in the loss-sharing, the rate is $\frac{t^- \alpha(A_h - A)}{r - (\alpha s_{gpls} - (1 - \bar{t})\alpha(A_h - A))}$.

screening is impaired, leading to a larger number of participants. They are ineffective on both intensive and extensive margins of saving. Finally, programs that combine concessional loans with loss- and profit-sharing in a suboptimal way could be doubly inefficient: they not only induce coordination when investment should be avoided (see $A'_{BF} < A_l$ in [Figure 5](#)), but also impose higher costs due to ineffective screening.

7 Conclusion

Political risk and coordination failure are considered two primary hindrances for growth in developing countries. We showed how political risk can induce coordination failure, and proposed subsidy programs to mitigate it at minimal cost compared to natural alternatives, including guarantees. The key is effective screening of potential investors and designing the program such that only the minimum necessary investors find it beneficial to take advantage of the program, while others invest, anticipating that the program encourages participation. We focused on political risk and development, but our approach can likewise be applied to encourage investment in specific locations, industries, or technologies whenever complementarities among investors exist and subsidies can be conditioned on investment decisions—see, e.g., [Gaubert, Hanson, and Neumark \(2024\)](#) on place-based incentive schemes.

Several directions for future research stand out:

1. An alternative approach to providing subsidy programs is that development agencies directly invest as large players. As [Corsetti et al. \(2004\)](#) show, even one large player can influence coordination. A natural question is whether and when international agencies can mitigate miscoordination at a lower cost by diverting funds from subsidies to direct investment.
2. Another direction is to study whether and how international agencies can provide information that facilitates coordinating investment ([Basak, Deb, and Kuvalekar, 2024](#); [Basak and Zhou, 2020](#)).

3. Our analysis was focused on settings with ex ante homogeneous investors (cf. [Luo and Yang, 2023](#); [Sákovics and Steiner, 2012](#)), in which there is little information asymmetry among investors. Analyzing the more general setting is left for future research.
4. We also restricted attention to settings where development agencies can observe investors' profits. In practice, investors may seek to conceal profits or overstate losses, while agencies can engage in costly monitoring. It remains to analyze how the optimal program is modified in the presence of reporting fraud and monitoring costs.
5. Our analysis focused on minimizing the costs of eliminating miscoordination inefficiencies. An alternative objective would be to examine what can be achieved through self-funded programs.
6. As an intermediate step toward implementing the proposed program in practice, lab experiments can provide guidance on its effectiveness. There is already a large experimental literature on different aspects of coordination and global games ([Avoyan, 2024](#)). Investigating subsidies in that context is a feasible and fruitful direction for future research.

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Online Appendix

A GPS and GPLS with Finite Noise

In [Proposition 4](#) and [Proposition 7](#) in the text, we characterize the equilibrium as the noise becomes vanishingly small. Here, we present propositions characterizing the equilibrium under a GPS and GPLS program for finite noise, $0 < \sigma < \min\{A_l - \underline{A}, \bar{A} - A_h\}$. The proofs of these propositions are in Online Appendix B.

For a fixed noise, $\sigma \in (0, \min\{A_l - \underline{A}, \bar{A} - A_h\})$, and a lower threshold x' , let $A_0(x', \sigma)$ denote the value of A at which investor i 's payoff is equal to zero, i.e., $\pi(K(A_0(x', \sigma); x'), A_0(x', \sigma)) = A_0(x', \sigma) - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A_0(x', \sigma); x'))} = 0$ where $K(A_0(x', \sigma); x') = \bar{K}Pr(x_i \geq x' | A_0(x', \sigma))$. Then,

$$A_0(x', \sigma) = \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + \bar{K}Pr(x_i \geq x' | A_0(x', \sigma)))}$$

Substituting the value of $Pr(x_i \geq x' | A_0(x', \sigma))$,

$$A_0(x', \sigma) = \begin{cases} A_h & ; x' > A_0(x', \sigma) + \sigma \\ \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + \frac{\bar{K}}{2} [1 - (\frac{x' - A_0(x', \sigma)}{\sigma})])} & ; x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma] \\ A_l & ; x' < A_0(x', \sigma) - \sigma \end{cases} \quad (15)$$

For $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$, solving for $A_0(x', \sigma)$ yields

$$A_0(x', \sigma) = \left(\frac{x'}{2} + \frac{r}{2\alpha} - \sigma \frac{\underline{K}}{\bar{K}} - \frac{\sigma}{2} \right) \pm \sqrt{\left(\frac{x'}{2} - \frac{r}{2\alpha} - \sigma \frac{\underline{K}}{\bar{K}} - \frac{\sigma}{2} \right)^2 + 2\sigma \frac{\delta}{\bar{K}(1-\alpha)^2}}.$$

$A_0(x', \sigma)$ is,

$$A_0(x', \sigma) = \left(\frac{x'}{2} + \frac{r}{2\alpha} - \sigma \frac{\underline{K}}{\bar{K}} - \frac{\sigma}{2} \right) + \sqrt{\left(\frac{x'}{2} - \frac{r}{2\alpha} - \sigma \frac{\underline{K}}{\bar{K}} - \frac{\sigma}{2} \right)^2 + 2\sigma \frac{\delta}{\bar{K}(1-\alpha)^2}}.$$

The other root yields $|A_0(x', \sigma) - x'| > \sigma$, which is not possible.

We can now present the equilibrium characterization. The following proposition states the result for a GPS program.

Proposition 10. *Suppose assumptions 1 and 2 hold and a GPS program (s, t) is in place. There is a unique equilibrium wherein $x' = x'(s, t) \leq x^*$, where*

$$x'(s, t) = \begin{cases} x^* - s & ; A_l \geq x^* - s + \sigma \\ \min\{\tilde{x}(s, t), x^*\} & ; A_l \leq x^* - s + \sigma, \end{cases}$$

and $\tilde{x} \in X' := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x < A_0(x, \sigma) + \sigma\}$ is the unique solution to

$$\tilde{x} = \frac{t}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1 - \alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)} \right) \right] + x^* - s.$$

Moreover,

- if $s \geq t(\bar{A} - A_l)$, then $x'' = \bar{A} + \sigma$, so that all who invest will participate in the program.
- if $s < t(\bar{A} - A_l)$, then $x'' = x''(s, t) = \max\{x'(s, t), x^\diamond(s, t)\}$, where $x^\diamond$ is the minimum x that solves

$$s = t \cdot E[\max\{\pi(K(A; x'), A), 0\} | x_i = x],$$

so that some who invest will not participate in the program.

The following proposition characterizes the equilibrium under a GPLS program when noise is finite.

Proposition 11. *Suppose assumptions 1 and 4 hold and a GPLS program (s, t^+, t^-) is in place. There is a unique equilibrium wherein $x' = x'(s, t^+, t^-) \leq x^*$, where*

$$x'(s, t^+, t^-) = \begin{cases} x^* - \frac{s}{1-t^-} & ; A_l \geq x^* - \frac{s}{1-t^-} + \sigma \\ \min\{\tilde{x}(s, t^+, t^-), x^*\} & ; A_l \leq x^* - \frac{s}{1-t^-} + \sigma, \end{cases}$$

where $\tilde{x} \in X' := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x < A_0(x, \sigma) + \sigma\}$ uniquely solves

$$\begin{aligned} \tilde{x} = & \frac{t^+}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)} \right) \right] \\ & + \frac{t^-}{2\sigma} \left[\frac{A_0(\tilde{x}, \sigma)^2 - (\tilde{x} - \sigma)^2}{2} - \frac{r(A_0(\tilde{x}, \sigma) - \tilde{x} + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)}{\underline{K}} \right) \right] \\ & + x^* - s. \end{aligned}$$

Moreover,

- if $s \geq t^+ (\bar{A} - A_l)$, then $x'' = \bar{A} + \sigma$, so that all who invest will participate in the program.
- if $s < t^+ (\bar{A} - A_l)$, then $x'' = x''(s, t^+, t^-) = \max\{x', x^\diamond(s, t^+, t^-)\}$, where $x^\diamond$ is the minimum x that solves

$$s = t^+ \cdot E[\max\{\pi(K(A; x'), A), 0\} | x_i = x] + t^- \cdot E[\min\{\pi(K(A; x'), A), 0\} | x_i = x],$$

so that some who invest will not participate in the program.

Online Appendix B: Proofs

Proof of Lemma 1. The first-order condition for the worker's problem (1) is $-(1-\alpha)Y + \sqrt{\delta Y/T} = 0$, i.e., $T = \frac{\delta}{(1-\alpha)^2} \frac{1}{Y} > 0$. The objective is strictly concave. Thus, the optimal tax rate $T^* = \min \left\{ \frac{\delta}{(1-\alpha)^2} \frac{1}{Y}, 1 \right\}$. $\square$

Proof of Proposition 1. Given the aggregate investment in the country is K , investor i invests in the country if and only if $\alpha A \left(1 - \frac{\delta}{(1-\alpha)^2 A(\underline{K} + \bar{K})} \right) - r \geq 0$. If $A < A_l$, then even if every other investor invests, i.e., $K = \bar{K}$, investor i does not invest because her returns from investing is $\alpha A \left(1 - \frac{\delta}{(1-\alpha)^2 A(\underline{K} + \bar{K})} \right) - r = \alpha(A - A_l) < 0$. Thus, not investing is the dominant strategy; in equilibrium, no investor invests when $A < A_l$. If $A \geq A_h$, even if no other investor invests ($K = 0$), investor i 's returns from investing in the country

is $\alpha A \left(1 - \frac{\delta}{(1-\alpha)^2 \bar{K}}\right) - r = \alpha(A - A_h) \geq 0$. Thus, investing is the dominant strategy; in equilibrium, all investors invest when $A \geq A_h$. For $A \in [A_l, A_h)$, both the equilibria - one where no one invests and the other where everyone invests, exist. $\square$

Proof of Proposition 4. For any GPS program (s, t) and for any $\sigma > 0$ that satisfies assumptions 1 and 2, Proposition 10 states that the unique equilibrium strategy is fully characterized by two cutoffs $(x'(s, t), x''(s, t))$: if investor i 's signal is $x_i < x'(s, t)$ then no investment, invest and participate if $x'(s, t) \leq x_i < x''(s, t)$ and invest without participating if $x_i \geq x''(s, t)$. Then, the existence of $\lim_{\sigma \rightarrow 0} x'(s, t)$ and $\lim_{\sigma \rightarrow 0} x''(s, t)$ implies that the equilibrium strategy converges to a two cutoff strategy with cutoffs $A'(s, t)$ and $A''(s, t)$ where $A'(s, t) = \lim_{\sigma \rightarrow 0} x'(s, t)$ and $A''(s, t) = \lim_{\sigma \rightarrow 0} x''(s, t)$. Therefore, it suffices to show $\lim_{\sigma \rightarrow 0} x'(s, t)$ and $\lim_{\sigma \rightarrow 0} x''(s, t)$ exist and characterize them.

For the lower cutoff x' , consider the following cases:

Case 1: $s \geq x^* - A_l$

From Proposition 10, $x' = x^* - s$ when $s \geq x^* - A_l + \sigma$. As the noise vanishes, for $s \geq x^* - A_l$,

$$\lim_{\sigma \rightarrow 0} x'(s, t) = A'(s, t) = x^* - s. \quad (16)$$

Case 2: $s \leq x^* - A_l$

From Proposition 10, for finite noise ($0 < \sigma < \min\{A_l - \underline{A}, \bar{A} - A_h\}$) and $s \leq x^* - A_l + \sigma$,

$\tilde{x}$ is the fixed point of $t \cdot H(x, \sigma) + x^* - s$ where

$$\begin{aligned} H(x, \sigma) &= \frac{1}{2\sigma} \left[\frac{(x + \sigma)^2 - A_0(x, \sigma)^2}{2} - \frac{r(x + \sigma - A_0(x, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma} \right) \right)} \right) \right] \\ &= \left(\frac{x - A_0(x, \sigma)}{2\sigma} + \frac{1}{2} \right) \left(\frac{x + A_0(x, \sigma) + \sigma}{2} - \frac{r}{\alpha} \right) - \frac{\delta}{(1-\alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma} \right) \right)} \right). \end{aligned} \quad (17)$$

As the noise becomes vanishingly small, $\lim_{\sigma \rightarrow 0} \tilde{x}$, if it exists, must solve

$$\lim_{\sigma \rightarrow 0} \tilde{x} = t \cdot h(\lim_{\sigma \rightarrow 0} \tilde{x}) + x^* - s,$$

where

$$h(x) = \lim_{\sigma \rightarrow 0} H(x, \sigma).$$

From [Proposition 10](#), $\tilde{x} \in X' := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x < A_0(x, \sigma) + \sigma\}$. Let X be the closure of the set X' , i.e., $X := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x \leq A_0(x, \sigma) + \sigma\}$. For $x \in X$,

$$h(x) = \lim_{\sigma \rightarrow 0} \left\{ \left(\frac{x - A_0(x, \sigma)}{2\sigma} + \frac{1}{2} \right) \left(\frac{x + A_0(x, \sigma) + \sigma}{2} - \frac{r}{\alpha} \right) - \frac{\delta}{(1-\alpha)^2 \bar{K}} \log \left(\frac{\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma} \right) \right)}}{\frac{\delta}{(1-\alpha)^2 \bar{K}}} \right) \right\}.$$

For $x \in X$, $\lim_{\sigma \rightarrow 0} A_0(x, \sigma) = x$. By L'Hopital's rule, $\lim_{\sigma \rightarrow 0} \frac{x - A_0(x, \sigma)}{\sigma} = \lim_{\sigma \rightarrow 0} -\frac{\partial}{\partial \sigma} A_0(x, \sigma) = 2\frac{\underline{K}}{\bar{K}} + 1 - \frac{2\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})}$. Then,

$$\begin{aligned} h(x) &= \left(\frac{\underline{K}}{\bar{K}} + 1 - \frac{\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} \right) \left(x - \frac{r}{\alpha} \right) - \frac{\delta}{(1-\alpha)^2 \bar{K}} \log \left(\frac{\frac{\underline{K} + \bar{K}}{(1-\alpha)^2 \bar{K}}}{\frac{\delta}{(1-\alpha)^2 \bar{K}}} \right) \\ &= \left(\frac{\underline{K}}{\bar{K}} + 1 \right) \left(x - \frac{r}{\alpha} \right) - \frac{\delta}{\bar{K}(1-\alpha)^2} \left(1 + \log \left(\frac{\underline{K} + \bar{K}}{\delta} \right) \right). \end{aligned} \quad (18)$$

We show the existence of $\lim_{\sigma \rightarrow 0} \tilde{x}$ and compute it by first showing that $H(x, \sigma)$ converges uniformly to $h(x)$ as $\sigma \rightarrow 0$. The following lemma states this.

Lemma 2. *For $x \in X$, $H(x, \sigma)$ converges uniformly to $h(x)$ as $\sigma \rightarrow 0$.*

Proof. Equation (18) demonstrates that $h(x) = \lim_{\sigma \rightarrow 0} H(x, \sigma)$ exists for any $x \in X$ and $\sigma < A_l - r/\alpha$ (note that $r/\alpha \notin X$). Also, $H(x, \sigma)$ is differentiable on X for $\sigma > 0$. We then prove that for $x \in X$, $\frac{\partial}{\partial x} H(x, \sigma)$ converges uniformly to a function $d(x)$. Then because $\lim_{\sigma \rightarrow 0} H(x, \sigma)$ exists for any $x \in X$, $H(x, \sigma)$ converges uniformly to $h(x)$ and $h'(x) = d(x)$ (Theorem 7.17 ([Rudin, 1976](#))).

For all $x \in X$, $H'(x, \sigma)$ exists and is

$$\begin{aligned} H'(x, \sigma) &= \frac{1}{2\sigma} \left[x + \sigma - A'_0(x, \sigma) \left(A_0(x, \sigma) - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2 \bar{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma} \right) \right)} \right) - A_0(x, \sigma) \right] \\ &= \frac{1}{2} \left(\frac{x - A_0(x, \sigma)}{\sigma} + 1 \right). \end{aligned}$$

Substituting the value of $\lim_{\sigma \rightarrow 0} \frac{x - A_0(x, \sigma)}{\sigma}$,

$$d(x) = \lim_{\sigma \rightarrow 0} H'(x, \sigma) = \left(\frac{\underline{K}}{\bar{K}} + 1 - \frac{\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} \right).$$

We show that for all $x \in X$, $H'(x, \sigma)$ converges uniformly to $d(x)$ as $\sigma \rightarrow 0$. Let $\sigma = 1/n$.

Denote $H'(x, 1/n)$ as $H'_n(x)$. Then,

$$\begin{aligned}
H'_n(x) - d(x) &= \frac{1}{2} \left[(n(x - A_0(x, 1/n)) + 1) - \left(2\frac{K}{\bar{K}} + 2 - \frac{2\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} \right) \right] \\
&= \frac{1}{2} \left[\frac{K}{\bar{K}} + \frac{1}{2} + \frac{nx}{2} - \frac{nr}{2\alpha} + 1 - \sqrt{\left(\frac{nx}{2} - \frac{nr}{2\alpha} - \frac{K}{\bar{K}} - \frac{1}{2} \right)^2 + \frac{2\delta n}{\bar{K}(1-\alpha)^2}} \right. \\
&\quad \left. - \left(2\frac{K}{\bar{K}} + 2 - \frac{2\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} \right) \right] \\
&= -\frac{1}{2} \left[\frac{K}{\bar{K}} + \frac{1}{2} - \frac{nx}{2} + \frac{nr}{2\alpha} + \sqrt{\left(\frac{nx}{2} - \frac{nr}{2\alpha} - \frac{K}{\bar{K}} - \frac{1}{2} \right)^2 + \frac{2\delta n}{\bar{K}(1-\alpha)^2}} - \frac{2\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} \right].
\end{aligned}$$

Denote $\frac{K}{\bar{K}} + \frac{1}{2} = k$ and $\frac{2\delta}{\bar{K}(1-\alpha)^2} = d$ and rewrite $x - \frac{r}{\alpha}$ as y . Then, for sufficiently large n ,

$$\begin{aligned}
|H'_n(y) - d(y)| &= \left| -\frac{1}{2} \left[\left(k - \frac{ny}{2} \right) + \sqrt{\left(\frac{ny}{2} - k \right)^2 + dn} - \frac{d}{y} \right] \right| \\
&= \left| -\frac{1}{2} \left[\left(k - \frac{ny}{2} \right) + \left(\frac{ny}{2} - k \right) \sqrt{1 + \frac{dn}{\left(\frac{ny}{2} - k \right)^2}} - \frac{d}{y} \right] \right|.
\end{aligned}$$

Using binomial series expansion to approximate the square root term,

$$\begin{aligned}
|H'_n(y) - d(y)| &= \left| \frac{1}{2} \left[\left(k - \frac{ny}{2} \right) \left(1 + \frac{dn}{2\left(\frac{ny}{2} - k \right)^2} \right) + O(1/n) - \left(k - \frac{ny}{2} \right) + \frac{d}{y} \right] \right| \\
&= \left| \frac{1}{2} \left[\frac{-d}{\left(y - \frac{2k}{n} \right)} + \frac{d}{y} + O(1/n) \right] \right|.
\end{aligned}$$

Then, for any $\epsilon > 0$, there exists an N such that $|H'_n(y) - d(y)| < \epsilon$ for all $n \geq N$ and for all y . This implies that for $x \in X$, $H'_n(x)$ converges uniformly to $d(x)$, i.e., $H'(x, \sigma)$ converges uniformly to $d(x)$ as $\sigma \rightarrow 0$. Furthermore, X is an interval $[A_l - \sigma, A_h + \sigma]$ because $x' \geq A_0(x', \sigma) - \sigma \implies x' \geq A_l - \sigma$. If that were not true, i.e., $x' < A_l - \sigma$, then, because $A_0(x', \sigma) \geq A_l$ (from equation (15)), $x' < A_0(x', \sigma) - \sigma$, which is a contradiction. By similar logic, $x' \leq A_0(x', \sigma) + \sigma \implies x' \leq A_h + \sigma$. Therefore $X = [A_l - \sigma, A_h + \sigma]$. Then, by Theorem 7.17 (Rudin, 1976), $H(x, \sigma)$ converges uniformly on X to $h(x)$ as $\sigma \rightarrow 0$, and $h'(x) = d(x)$ for $x \in X$. A direct calculation verifies that $h'(x) = d(x)$ indeed holds. $\square$

For $x \in X'$, the slope of $h(x)$ is

$$\frac{\partial}{\partial x} h(x) = \left(\frac{K}{\bar{K}} - \frac{\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} + 1 \right).$$

For all $x \in X'$, $\lim_{\sigma \rightarrow 0} \frac{x - A_0(x, \sigma)}{\sigma} \in [-1, 1)$. Then $\lim_{\sigma \rightarrow 0} \frac{x - A_0(x, \sigma)}{\sigma} = 2\frac{K}{\bar{K}} + 1 - \frac{2\delta}{\bar{K}(1-\alpha)^2(x - \frac{r}{\alpha})} < 1 \implies$

$\left(\frac{K}{\bar{K}} - \frac{\delta}{\bar{K}(1-\alpha)^2(x-\frac{r}{\alpha})} + 1\right) < 1$. The slope of $h(x)$ is $\frac{\partial}{\partial x}h(x) < 1$. Therefore, $h(x)$ admits a unique fixed point and because $t \in [0, 1]$, $t \cdot h(x) + x^* - s$ admits a unique fixed point. Denote it by $\tilde{A}$.

Next, we show that $\lim_{\sigma \rightarrow 0} \tilde{x}$ exists and is equal to $\tilde{A}$. From [Proposition 10](#), for any $\sigma \in (0, \min\{A_l - \underline{A}, \bar{A} - A_h\})$, there exists a unique fixed point $\tilde{x}$ (denote it by $\tilde{x}(\sigma)$) that solves

$$\tilde{x}(\sigma) = t \cdot H(\tilde{x}(\sigma), \sigma) + x^* - s.$$

Because we define $H(x, \sigma)$ on $X' \subset [A_l - \sigma, A_h + \sigma]$, the fixed point $\tilde{x}(\sigma) \in [A_l - \sigma, A_h + \sigma]$. Also, $H(x, \sigma)$ is continuous in both x and σ . Then, existence of $\lim_{\sigma \rightarrow 0} \tilde{x}(\sigma)$ follows because $h(x)$ admits a unique fixed point. We showed earlier that $t \cdot h(x) + x^* - s$ admits a unique fixed point $\tilde{A}$. Therefore, it must be that $\lim_{\sigma \rightarrow 0} \tilde{x}(\sigma) = \lim_{\sigma \rightarrow 0} \tilde{x} = \tilde{A}$.

Substituting $\tilde{A}$ for x for $h(\tilde{A})$ (equation (18)), when $s \leq x^* - A_l$, as the noise becomes vanishingly small, $\lim_{\sigma \rightarrow 0} \tilde{x} = \tilde{A}$ uniquely solves,

$$\tilde{A} = x^* - s + t \frac{\delta}{(1-\alpha)^2 \bar{K}} \left\{ \frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} - \log \left(\frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} \right) - 1 \right\}. \quad (19)$$

Note that there exist s, t such that $\tilde{A} > x^*$ (for example, $s = 0$ & $t = 1$ yields $\tilde{A} > x^*$). However, from [Proposition 10](#), $x'(s, t) = \min\{\tilde{x}(s, t), x^*\}$ (the lower cutoff will never be greater than the cutoff without the program, x^*). Thus, when $s \leq x^* - A_l$, in the limit, the lower cutoff is

$$A'(s, t) = \lim_{\sigma \rightarrow 0} x'(s, t) = \min\{\tilde{A}(s, t), x^*\}. \quad (20)$$

Next, we compute the upper cutoff x'' . For a small noise ($0 < \sigma < \min\{A_l - \underline{A}, \bar{A} - A_h\}$), when $s/t \geq \bar{A} - A_l = \pi(\bar{K}, \bar{A})$, $t\pi(\bar{K}, \bar{A}) - s = \pi(2, \bar{K}, \bar{A}) - \pi(1, \bar{K}, \bar{A}) \leq 0$. Therefore, the upper cutoff is $x'' = \bar{A} + \sigma$. As the noise becomes vanishingly small, $x'' = \bar{A}$ in this case.

Finally, consider the case where $s/t < \bar{A} - A_l = \pi(\bar{K}, \bar{A})$. As we show in [Proposition 10](#), the upper cutoff is $x''(s, t) = \max\{x'(s, t), x^\diamond(s, t)\}$, where $x^\diamond$ is the minimum x that solves

$$\frac{s}{t} = E[\max\{\pi(K(A; x'), A), 0\} | x_i = x]. \quad (21)$$

For $0 < \sigma < \min\{A_l - \underline{A}, \bar{A} - A_h\}$, the two cutoffs $x''(s, t)$ and $x'(s, t)$ induced by the GPS program (s, t) must satisfy either $x''(s, t) > x'(s, t) + 2\sigma$ or $x''(s, t) \leq x'(s, t) + 2\sigma$. Suppose,

it is the first case, i.e., $x''(s, t) > x'(s, t) + 2\sigma$. Then, in this case, from equation (21),

$$\begin{aligned}\frac{s}{t} &= E\left[\max\left\{\left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right), 0\right\} \middle| x_i = x''\right] \\ \frac{s}{t} &= \int_{x''-\sigma}^{x''+\sigma} \left(A - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \overline{K}}\right) f(A|x_i = x'') dA \\ x'' &= \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \overline{K}} + \frac{s}{t} = A_l + \frac{s}{t}\end{aligned}$$

which is σ -independent.

We showed above that $\lim_{\sigma \rightarrow 0} x'(s, t) = A'(s, t)$ exists. Then, if $A'(s, t) < A_l + \frac{s}{t}$, for sufficiently small σ , $x'(s, t) + 3\sigma < A_l + \frac{s}{t}$. Also, in this case, it must be that $s/t > 0$. Then, from equation (21), note that $x^\diamond(s, t) \geq A_l + \frac{s}{t} - \sigma$. This is because $E[\max\{\pi(K(A; x'), A), 0\} | x_i = x^\diamond] \leq E[\max\{\pi(K(A; x'), A), 0\} | x_i = x^\diamond] \leq E[\max\{A - A_l, 0\} | x_i = x^\diamond] \leq x^\diamond + \sigma - A_l$. Therefore, from equation (21), $s/t \leq x^\diamond(s, t) + \sigma - A_l \implies x^\diamond(s, t) \geq A_l + \frac{s}{t} - \sigma$. Then, $x'(s, t) + 2\sigma < x^\diamond(s, t)$. Because $x''(s, t) = \max\{x'(s, t), x^\diamond(s, t)\}$, $x'(s, t) + 2\sigma < x''(s, t)$. We showed above that for $x'(s, t) + 2\sigma < x''(s, t)$, $x''(s, t) = A_l + \frac{s}{t}$. Thus, if $A'(s, t) < A_l + \frac{s}{t}$, for any sufficiently small σ , $x''(s, t) = A_l + \frac{s}{t}$. Therefore, in this case,

$$\lim_{\sigma \rightarrow 0} x''(s, t) = A''(s, t) = \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2(\underline{K} + \overline{K})} + \frac{s}{t}. \quad (22)$$

If $A'(s, t) > A_l + \frac{s}{t}$, then for sufficiently small σ , $x'(s, t) + 2\sigma > A_l + \frac{s}{t}$. Then, it must be that for any sufficiently small σ , $x'(s, t) + 2\sigma \geq x''(s, t)$. Because, if $x'(s, t) + 2\sigma < x''(s, t)$, $x''(s, t) = A_l + \frac{s}{t} \implies x'(s, t) + 2\sigma < A_l + \frac{s}{t}$ which is a contradiction. Thus, in this case, for any arbitrarily small σ , $x''(s, t) \leq x'(s, t) + 2\sigma \implies \lim_{\sigma \rightarrow 0} x''(s, t) = \lim_{\sigma \rightarrow 0} x'(s, t)$. Then,

$$A''(s, t) = A'(s, t). \quad (23)$$

Finally, when $A'(s, t) = A_l + \frac{s}{t}$, both candidate limits for $x''(s, t)$ coincide so that $\lim_{\sigma \rightarrow 0} x''(s, t) = A''(s, t) = \lim_{\sigma \rightarrow 0} x'(s, t) = A'(s, t) = A_l + \frac{s}{t}$.

Thus, for any GPS program (s, t) , A'' exists and can be characterized by either equation (22) or equation (23), depending on the program parameters s and t . When $A''(s, t) > A'(s, t)$, its value is given by equation (22), which is continuous and increasing in s for a given t , whereas $A'(s, t)$ is continuous and decreasing in s , reaching $A'(s, t) = x^*$. Consequently,

by the intermediate value theorem, for any given t , there exists an s such that $A''(s, t) = A_l + \frac{s}{t} = A'(s, t)$. Therefore, as the noise becomes vanishingly small, the upper cutoff $A''(s, t)$ is

$$A''(s, t) = \begin{cases} A_l + s/t & ; s \geq t (A'(s, t) - A_l) \\ A'(s, t) & ; s \leq t (A'(s, t) - A_l). \end{cases}$$

This concludes the proof. □

Proof of Proposition 5. The proof consists of three parts:

1. Show that for a given t and a target x' , there is a unique guarantee $s(x', t)$ that induces x' .
2. Show that the cost of implementing any given target x' is decreasing t .
3. Compute the expected costs for removing the inefficiency interval with the optimal GPS program as the noise becomes vanishingly small.

The following lemma proves part 1,

Lemma 3. *Under assumptions 1 and 2, for any $x' \in (x^* - \bar{s}, x^*)$, the set of GPS programs $\{(s, t) : s \in [0, \bar{s}], t \in [0, 1]\}$ that implement the target x' is non-empty. Furthermore, for a given t , if x' is implementable with that t , then there is a unique s , denoted by $s(x', t)$, that implements x' .*

Proof. Any $x' \in (x^* - \bar{s}, x^*)$ will either satisfy $x' < A_0(x', \sigma) - \sigma$ or $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma)$. This follows because any x' satisfies $x' < A_0(x', \sigma) + \sigma$. If $x' \geq A_0(x', \sigma) + \sigma$, then, from equation (15), $x' \geq A_h + \sigma \implies x' > A_h$ ($\sigma > 0$), which is a contradiction because $x' \leq x^* \leq A_h$.

Then, consider the following two cases:

Case 1: $x' < A_0(x', \sigma) - \sigma$

From equation (15), $A_0(x', \sigma) = A_l$. In this case, $x' < A_l - \sigma$. It follows from Proposition 10 that any (s, t) that satisfies the following equation implements x' ,

$$x' = x'(s, t) = \frac{r}{\alpha} + \frac{\delta}{(1 - \alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - s.$$

Then, the set of programs that achieves x' when $x' < A_0(x', \sigma) - \sigma$ is $\{(s, t) : t \in [0, 1], s = x^* - x'\}$. Therefore, for any given t , there exists a unique guarantee $s(x', t)$ that achieves x' where,

$$s(x', t) = x^* - x'. \quad (24)$$

Case 2: $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$

For $x' < x^*$, any (s, t) that satisfies the following implements x' (from [Proposition 10](#)), $x' - T(x') = 0$, where

$$T(x') = \frac{t}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} - \frac{2\sigma\delta}{(1 - \alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right) \right] + x^* - s.$$

For any $t \in [0, 1]$, $T'(x') = \frac{t}{2\sigma}(x' + \sigma - A_0(x', \sigma)) \in [0, 1]$. Then, by the implicit function formula, $\frac{dx'}{ds} = -\frac{1}{1 - T'(x')} < 0$. This implies that for a given t , there is a one-to-one mapping from x' to s . Thus, there exists a set of programs (s, t) that achieves any $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$ and for a given t , there is a unique guarantee $s(x', t)$ that achieves x' . Furthermore, for $x' = A_0(x', \sigma) - \sigma$, for a given t , the guarantee $s(x', t)$ that achieves it $s(x', t) = x^* - x'$.

Therefore, for any $x' \in (x^* - \bar{s}, x^*)$, the set of programs that achieves the target x' is non-empty and for a given t , whenever that x' is implementable by t , there is a unique $s(x', t)$ that achieves x' . $\square$

The following lemma proves the second part:

Lemma 4. *Under assumptions [1](#) and [2](#), the expected cost to the policymaker to implement a target x' , where $x' \in (x^* - \bar{s}, x^*)$, is (weakly) decreasing in t .*

Proof. Following [Lemma 3](#), we can now state the GPS program (s, t) in terms of the profit sharing rate t and x' that it induces, i.e., (x', t) . Let $C(A, x', t; \sigma)$ denote the cost of inducing target x' for a given t and A . The expected cost of the program (x', t) is,

$$\begin{aligned}
E_A[C(A, x', t; \sigma)] &= \int_{\underline{A}}^{\bar{A}} (s(x', t) - t \cdot \max\{\pi(K(A; x'), A), 0\}) \cdot (F(x''|A) - F(x'|A)) f(A) dA \\
&= \int_{\underline{A}}^{\bar{A}} [\pi(1, K(A; x'), A) - \pi(2, K(A; x'), A)] (F(x''|A) - F(x'|A)) f(A) dA \\
&= \int_{\underline{A}}^{\bar{A}} [\pi(1, K(A; x'), A) - \pi(2, K(A; x'), A)] \left(\int_{x'}^{x''} f(x_i|A) dx_i \right) f(A) dA \\
&= \int_{\underline{A}}^{\bar{A}} \left(\int_{x'}^{x''} [\pi(1, K(A; x'), A) - \pi(2, K(A; x'), A)] f(x_i|A) dx_i \right) f(A) dA \\
&= \int_{x'}^{x''} \int_{\underline{A}}^{\bar{A}} [\pi(1, K(A; x'), A) - \pi(2, K(A; x'), A)] f(A|x_i) dA f(x_i) dx_i \\
&= \int_{x'}^{x''} \left(E[\pi(1, K(A; x'), A)|x_i] - E[\pi(2, K(A; x'), A)|x_i] \right) f(x_i) dx_i \\
&= \int_{x'}^{\bar{A}+\sigma} \max\{E[\pi(1, K(A; x'), A) - \pi(2, K(A; x'), A)|x_i], 0\} f(x_i) dx_i. \quad (25)
\end{aligned}$$

Because $E[\pi(2, K(A; x'), A)|x_i]$ does not depend on t , the relationship between the expected cost of implementing x' , $E_A[C(A, x', t; \sigma)]$, and t depends on how $E[\pi(1, K(A; x'), A)|x_i]$ varies with t . Then,

$$\begin{aligned}
E[\pi(1, K(A; x'), A)|x_i] &= \int_{\underline{A}}^{\bar{A}} (\pi(K(A; x'), A) - t \max\{\pi(K(A; x'), A), 0\} + s) f(A|x_i) dA \\
&= \int_{\underline{A}}^{\bar{A}} \pi(2, K(A; x'), A) f(A|x_i) dA - t \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + s \\
&= E[\pi(2, K(A; x'), A)|x_i] - t \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + s,
\end{aligned}$$

where $s = s(x', t)$. Then for a fixed x' ,

$$\frac{d}{dt} E[\pi(1, K(A; x'), A)|x_i] = - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + \frac{ds(x', t)}{dt}.$$

Next, as we do in lemma 3, consider the two cases:

Case 1: $x' < A_0(x', \sigma) - \sigma$

In this case, from equation (24), $s(x', t) = x^* - x'$. Then, for a fixed x' , $\frac{ds(x', t)}{dt} = 0$ and $\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA \geq 0$ for $x_i > x'$. Then, $\frac{d}{dt} E[\pi(1, K(A; x'), A)|x_i] \leq 0$ for all $x_i > x'$.

Case 2: $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$

For any given x' , the marginal investor is indifferent between investing with participation and not investing which gives us the relationship between $s(x', t)$ & t .

$$s(x', t) = t \cdot E[\max\{\pi(K(A; x'), A), 0\} | x_i = x'] - E[\pi(K(A; x'), A) | x_i = x']$$

$$\frac{d}{dt}s(x', t) = E[\max\{\pi(K(A; x'), A), 0\} | x_i = x'] = \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x') dA.$$

In this case,

$$\begin{aligned} \frac{d}{dt}E[\pi(1, K(A; x'), A) | x_i] &= - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x_i) dA + \frac{ds(x', t)}{dt} \\ &= - \left(\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x_i) dA - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x') dA \right). \end{aligned}$$

For $A > A_0(x', \sigma)$, $\max\{\pi(K(A; x'), A), 0\} > 0$ and is strictly increasing in A . Therefore, $\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x_i) dA$ is strictly increasing in x_i for $x_i > A_0(x', \sigma) - \sigma$. Because $x' \geq A_0(x', \sigma) - \sigma$, $\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x_i) dA$ is strictly increasing in x_i for $x_i > x'$. Furthermore, we only need to check for $x_i > x'$ since that is the range over which we integrate in equation (25). Therefore, for any $x_i > x'$,

$$\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x_i) dA > \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A | x') dA.$$

This implies that $\frac{d}{dt}E[\pi(1, K(A; x'), A) | x_i] < 0$. Therefore, in this case, the expected cost of implementing x' is strictly decreasing in t . $\square$

Finally, we compute the expected cost of removing miscoordination as the noise becomes vanishingly small. To remove inefficiency, the target must be $x' = A_l - \sigma$. Assumptions 2 and 3 ensure that it is feasible to achieve this target, i.e., $x^* - A_l + \sigma \leq \bar{s} < A_l - \underline{A} - \sigma$. Because the cost is decreasing in t , we can set the highest profit-sharing rate, $t = 1$. From Proposition 10, the corresponding compensation, $s(x', 1)$, is $s(x', 1) = \frac{\delta}{(1-\alpha)^2 \bar{K}} \left[\log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{\bar{K}}{\underline{K} + \bar{K}} \right] + \sigma = x^* - A_l + \sigma$. Given $s(x', 1)$ and $x' = A_l - \sigma$, it follows that the upper cutoff is $x'' > A_l + \sigma$. Then, computing the upper cutoff x'' (from Proposition 10) yields

$$x'' = A_l + s(x', 1) = \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2 \overline{K}} \log \left(\frac{\underline{K} + \overline{K}}{\underline{K}} \right) + \sigma = x^* + \sigma. \quad (26)$$

Thus, the ex-ante cost is

$$\begin{aligned} E_A[C(A, x', t = 1; \sigma)] &= \int_{\underline{A}}^{\overline{A}} \left((s(x', 1) - 1 \cdot \max\{\pi(K(A; x'), A), 0\}) \cdot (F(x''|A) - F(x'|A)) \right) f(A) dA \\ &= 0 + \int_{x'-\sigma}^{x'+\sigma} \left((s(x', 1) - \max\{\pi(K(A; x'), A), 0\}) \cdot \left(1 - \frac{1}{2} \left(\frac{x' - A}{\sigma} + 1 \right) \right) \right) f(A) dA \\ &\quad + \int_{x'+\sigma}^{x''-\sigma} \left((s(x', 1) - \max\{\pi(K(A; x'), A), 0\}) \cdot (1 - 0) \right) f(A) dA \\ &\quad + \int_{x''-\sigma}^{x''+\sigma} \left((s(x', 1) - \max\{\pi(K(A; x'), A), 0\}) \cdot \left(\frac{1}{2} \left(\frac{x'' - A}{\sigma} + 1 \right) - 0 \right) \right) f(A) dA + 0 \\ &= \int_{x'-\sigma}^{x'+\sigma} \frac{s(x', 1)}{2} \left(1 - \left(\frac{x' - A}{\sigma} \right) \right) f(A) dA + \int_{x'+\sigma}^{x''-\sigma} (s(x', 1) - \pi(K(A; x'), A)) f(A) dA \\ &\quad + \int_{x''-\sigma}^{x''+\sigma} \left(\frac{(s(x', 1) - \pi(K(A; x'), A))}{2} \left(\frac{x'' - A}{\sigma} + 1 \right) \right) f(A) dA \\ &= \frac{1}{\overline{A} - \underline{A}} \left\{ s(x', 1) \sigma + (x'' - x' - 2\sigma) \left[s(x', 1) - \left(\frac{x'' + x'}{2} - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2 (\underline{K} + \overline{K})} \right) \right] \right. \\ &\quad \left. + \sigma \left[s(x', 1) + \left(\frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2 (\underline{K} + \overline{K})} \right) + \frac{\sigma}{3} - x'' \right] \right\} \end{aligned}$$

As the noise vanishes, the cost converges to

$$\begin{aligned} \lim_{\sigma \rightarrow 0} E_A[C(A, x', 1; \sigma)] &= \lim_{\sigma \rightarrow 0} \left\{ \frac{(x'' - x')}{\overline{A} - \underline{A}} \left[s(x', 1) - \left(\frac{x'' + x'}{2} - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2 (\underline{K} + \overline{K})} \right) \right] \right\} \\ &= \lim_{\sigma \rightarrow 0} \left\{ \frac{(x'' - x')}{\overline{A} - \underline{A}} \left[s(x', 1) - \left(\frac{x'' + x'}{2} - A_l \right) \right] \right\}. \end{aligned}$$

Substituting $\lim_{\sigma \rightarrow 0} s(x', 1) = x^* - A_l$, $x' = A'(\lim_{\sigma \rightarrow 0} s(x', 1), t) = A_l$, and $x'' = A''(\lim_{\sigma \rightarrow 0} s(x', 1), 1) = A_l + s(x', 1) = x^*$ (from proposition 4),

$$\lim_{\sigma \rightarrow 0} E_A[C(A, x', 1; \sigma)] = \lim_{\sigma \rightarrow 0} \left\{ \frac{(x^* - A_l)}{\overline{A} - \underline{A}} \left[x^* - A_l - \left(\frac{x^* + A_l}{2} - A_l \right) \right] \right\} = \frac{(x^* - A_l)^2}{2(\overline{A} - \underline{A})}.$$

Therefore, as the noise becomes vanishingly small, the optimal GPS program that removes miscoordination is $(s, t) = (x^* - A_l, 1)$, and its expected cost is $\frac{1}{2} \frac{(x^* - A_l)^2}{\overline{A} - \underline{A}}$. $\square$

Proof of Proposition 6. We begin by proving a lemma.

Lemma 5. Let $G(k) = \frac{1}{k} \int_0^k (g(x) - g(k)) dx$, where $k > 0$ and $g(x) : [0, \infty) \rightarrow (0, \infty)$ is a twice continuously differentiable function with $g'(x) < 0$.

1. $\lim_{k \rightarrow 0} G'(k) > 0$.
2. If $\lim_{x \rightarrow \infty} x g'(x) = 0$, then $G'(k) < 0$ for sufficiently large k .
3. If $\lim_{x \rightarrow \infty} x g'(x) = 0$ and the solution to $G(k) = -k g'(k)$ is unique, then $G(k)$ is strictly single-peaked.
4. If $\lim_{x \rightarrow \infty} x g'(x) = 0$ and $-x g''(x)/g'(x) > 3/2$, then $((G(k))^2 k)' < 0$ for sufficiently large k .

Proof of Lemma 5. Differentiating $G(k)$ yields

$$G'(k) = -\frac{1}{k^2} \int_0^k (g(x) - g(k)) dx - g'(k) = -\frac{G(k)}{k} - g'(k). \quad (27)$$

Part 1. From (27),

$$-\lim_{k \rightarrow 0} G'(k) = \lim_{k \rightarrow 0} \frac{G(k)}{k} + \lim_{k \rightarrow 0} g'(k) = \lim_{k \rightarrow 0} G'(k) + \lim_{k \rightarrow 0} g'(k) \Rightarrow \lim_{k \rightarrow 0} G'(k) = -\lim_{k \rightarrow 0} g'(k)/2 > 0,$$

where the second equality follow from L'Hopital's rule and the last inequality follows from $g'(k) < 0$ for all k .

Part 2. From (27), $G'(k) < 0$ if and only if $-k g'(k) < G(k)$. Further, integration by parts yields

$$G(k) = \frac{1}{k} \int_0^k -x g'(x) dx.$$

Because $\lim_{x \rightarrow 0} -x g'(x) = 0$, for sufficiently large K and $k > K$, we have

$$k G(k) \approx \int_0^K -x g'(x) dx + (k - K)(-k g'(k))$$

Thus, it suffices to show

$$-k k g'(k) < \int_0^K -x g'(x) dx + (k - K)(-k g'(k)).$$

That is,

$$0 < \int_0^K -x g'(x) dx + K k g'(k) \Leftrightarrow -k g'(k) < \frac{1}{K} \int_0^K -x g'(x) dx.$$

The result follows because $-k g'(k) \approx 0$ for sufficiently large k .

Part 3. By parts 1 and 2, $G(k)$ is strictly increasing for small k and strictly decreasing for large k . Thus, it has at least a local maximum, at which $G'(x) = 0$. Thus, it suffices to show that $G'(k) = -G(k)/k - g'(k) = 0$ has no other solution, where the first equality is from (27).

Part 4. $((G(k))^2 k)' = G^2(2kG'/G + 1) = G^2(-2 - 2kg'/G + 1) = -G^2(1 + 2kg'/G)$, where the second equality follows from (27). It suffices to show that, for sufficiently large k ,

$$\frac{k g'}{G} > -\frac{1}{2}. \quad (28)$$

If $\lim_{x \rightarrow \infty} G(x) > 0$, then the result follows because $-k g'(k) \approx 0$ for sufficiently large k .

Next, suppose $\lim_{x \rightarrow \infty} G(x) = 0$. By L'Hopital's rule we have

$$\lim_{k \rightarrow \infty} \frac{k g'(k)}{G(k)} = \lim_{k \rightarrow \infty} -\frac{g'(k) + k g''(k)}{G(k)/k + g'(k)} = \lim_{k \rightarrow \infty} -\frac{1 + \frac{k g''(k)}{g'(k)}}{1 + \frac{G(k)}{k g'(k)}},$$

where the first equality follows from (27). Thus,

$$\lim_{k \rightarrow \infty} \frac{k g'(k)}{G(k)} + 1 = -1 - \lim_{k \rightarrow \infty} \frac{k g''(k)}{g'(k)} \Leftrightarrow \lim_{k \rightarrow \infty} \frac{k g'(k)}{G(k)} = -2 - \lim_{k \rightarrow \infty} \frac{k g''(k)}{g'(k)}.$$

Thus,

$$\lim_{k \rightarrow \infty} \frac{k g'}{G} > -\frac{1}{2} \Leftrightarrow -2 - \lim_{k \rightarrow \infty} \frac{k g''(k)}{g'(k)} > -\frac{1}{2} \Leftrightarrow \lim_{k \rightarrow \infty} -\frac{k g''(k)}{g'(k)} > \frac{3}{2}.$$

The result follows from (28) and the assumption. □

From [Proposition 5](#), the optimal GPS transfer that eliminates miscoordination is $s = (x^* - A_l)$. Denote the function

$$G(\bar{K}) = \frac{(1-\alpha)^2}{\delta} (x^*(\bar{K}) - A_l(\bar{K})) = \frac{1}{\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{1}{\underline{K} + \bar{K}} = \frac{1}{\bar{K}} \int_0^{\bar{K}} (g(K) - g(\bar{K})) dK,$$

where $g(K) = \frac{1}{\underline{K} + K}$. $g(K)$ is twice continuously differentiable for $K \geq 0$ and $g'(\bar{K}) = -\frac{1}{(\underline{K} + \bar{K})^2} < 0$ for all $\bar{K} \geq 0$. Then, from part 1 in [Lemma 5](#), $\lim_{\bar{K} \rightarrow 0} G'(\bar{K}) > 0$.

Next,

$$\lim_{\bar{K} \rightarrow \infty} \bar{K} g'(\bar{K}) = - \lim_{\bar{K} \rightarrow \infty} \frac{\bar{K}}{(\underline{K} + \bar{K})^2} = 0.$$

Therefore, from part 2 in [Lemma 5](#), $G'(\bar{K}) < 0$ for sufficiently large $\bar{K}$.

It follows from part 3 in [Lemma 5](#) that $G(\bar{K}) = \frac{1}{\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{1}{\underline{K} + \bar{K}}$ is strictly single-peaked if there is a unique $\bar{K}$ that solves

$$\begin{aligned} \frac{1}{\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{1}{\underline{K} + \bar{K}} &= \frac{\bar{K}}{(\underline{K} + \bar{K})^2} \\ \frac{\bar{K}}{\underline{K} + \bar{K}} \left[1 + \frac{\bar{K}}{\underline{K} + \bar{K}} \right] - \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) &= 0 \end{aligned} \quad (29)$$

Existence of a solution to equation (29) follows because, as we showed above, $\lim_{\bar{K} \rightarrow 0} G'(\bar{K}) > 0$ and $G'(\bar{K}) < 0$ for sufficiently large $\bar{K}$.

To show uniqueness, let $R(\bar{K}) = \frac{\bar{K}}{\underline{K} + \bar{K}} \left[1 + \frac{\bar{K}}{\underline{K} + \bar{K}} \right]$, $L(\bar{K}) = \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right)$, and $D(\bar{K}) = R(\bar{K}) - L(\bar{K})$. Both $L(\bar{K})$ and $R(\bar{K})$ are continuous and differentiable in $\bar{K}$ for $\bar{K} \in (0, \infty)$. Thus, $\lim_{\bar{K} \rightarrow 0} D(\bar{K}) = \lim_{\bar{K} \rightarrow 0} (R(\bar{K}) - L(\bar{K})) = 0$.

Because $D'(\bar{K}) = R'(\bar{K}) - L'(\bar{K}) = \frac{\bar{K}(\underline{K} - \bar{K})}{(\underline{K} + \bar{K})^3}$, $D'(\bar{K}) > 0$ for $\bar{K} \in (0, \underline{K})$ and $D'(\bar{K}) < 0$ for $\bar{K} \in (\underline{K}, \infty)$. Thus, $D(\bar{K})$ is positive and increasing on $(0, \underline{K})$, therefore, $D(\bar{K}) = 0$ has no solution on $(0, \bar{K}]$. Then, any solution must lie in $(\underline{K}, \infty)$. $D(\bar{K})$ is strictly decreasing on $(\underline{K}, \infty)$ because $D'(\bar{K}) < 0$ for $\bar{K} \in (\underline{K}, \infty)$. Hence, equation (29) has a unique solution on $(0, \infty)$. Thus, from part 3 in [Lemma 5](#), $G(\bar{K})$ is strictly single-peaked.

From [Proposition 5](#), the cost (per unit of capital) of the optimal GPS program is

$$C(GPS) = \frac{(s^*)^2}{2(\bar{A} - \underline{A})} = \left(\frac{\delta}{(1 - \alpha)^2} \right)^2 \frac{G^2(\bar{K})}{2(\bar{A} - \underline{A})}.$$

Because $G(\bar{K}) > 0$ for $\bar{K} \in (0, \infty)$ and is strictly single-peaked, $G^2(\bar{K})$ is also strictly single-peaked. Therefore, $C(GPS)$ is strictly single-peaked in $\bar{K}$.

The total cost is proportional to $G^2(\bar{K}) \bar{K}$, and $-\frac{\bar{K} g''(\bar{K})}{g'(\bar{K})} = -\frac{\bar{K} \frac{2}{(\underline{K} + \bar{K})^3}}{-\frac{1}{(\underline{K} + \bar{K})^2}} = 2 \frac{\bar{K}}{\underline{K} + \bar{K}}$ approaches 2 for sufficiently large $\bar{K}$. Thus, by Part 4 of [Lemma 5](#), $G^2(\bar{K}) \bar{K}$ is decreasing for sufficiently large $\bar{K}$. $\square$

Proof of Proposition 7. The proof follows a similar approach to that of the proof for [Proposition 4](#). As in [Proposition 10](#), it suffices to show $\lim_{\sigma \rightarrow 0} x'(s, t^+, t^-)$ and $\lim_{\sigma \rightarrow 0} x''(s, t^+, t^-)$ exist and characterize them. Consider the following cases for x' :

Case 1: $\frac{s}{1-t^-} \geq x^* - A_l$

For $0 < \sigma < \min\{A_l - \underline{A}, \bar{A} - A_h\}$, $x' = x^* - \frac{s}{1-t^-}$ when $\frac{s}{1-t^-} \geq x^* - A_l + \sigma$ (from [Proposition 11](#)). As the noise becomes vanishingly small, for $\frac{s}{1-t^-} \geq x^* - A_l$,

$$x' = A'(s, t^+, t^-) = x^* - \frac{s}{1-t^-}. \quad (30)$$

Case 2: $\frac{s}{1-t^-} \leq x^* - A_l$

From [Proposition 11](#), the cutoff is $x' = \min\{\tilde{x}(s, t^+, t^-), x^*\}$ where $\tilde{x}$ uniquely solves

$$\begin{aligned} \tilde{x} = & \frac{t^+}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)} \right) \right] \\ & + \frac{t^-}{2\sigma} \left[\frac{A_0(\tilde{x}, \sigma)^2 - (\tilde{x} - \sigma)^2}{2} - \frac{r(A_0(\tilde{x}, \sigma) - \tilde{x} + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)}{\underline{K}} \right) \right] \\ & + x^* - s. \end{aligned}$$

Let $t^+ = t^- + \gamma$. We can rewrite the implicit function for $\tilde{x}$ as

$$\begin{aligned}
\tilde{x} &= \frac{\gamma}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)} \right) \right] \\
&\quad + \frac{t^-}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - (\tilde{x} - \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - \tilde{x} + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) \right] \\
&\quad + x^* - s \\
\tilde{x} &= \frac{\gamma}{1-t^-} \frac{1}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)} \right) \right] \\
&\quad + x^* - \frac{s}{1-t^-}.
\end{aligned}$$

Therefore, in the GPLS case, $\tilde{x}$ is the fixed point of the function $\frac{\gamma}{1-t^-}H(x, \sigma) + x^* - \frac{s}{1-t^-}$ where $H(x, \sigma)$ is given by equation (17). For $x \in X = \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x \leq A_0(x, \sigma) + \sigma\}$, $H(x, \sigma)$ converges uniformly to $h(x)$ (Lemma 2). Because $\frac{\gamma}{1-t^-} = \frac{t^+ - t^-}{1-t^-} \leq 1$, $\frac{\gamma}{1-t^-}h(x) + x^* - \frac{s}{1-t^-}$ admits a unique fixed point, $\tilde{A}$. Then, using the same arguments as in the proof of Proposition 4, $\lim_{\sigma \rightarrow 0} \tilde{x} = \tilde{A}$.

Calculating $h(\tilde{A})$ (equation (18)), when $\frac{s}{1-t^-} \leq x^* - A_l$, as the noise becomes vanishingly small, $\lim_{\sigma \rightarrow 0} \tilde{x} = \tilde{A}$ uniquely solves

$$\tilde{A} = x^* - \frac{s}{1-t^-} + \frac{t^+ - t^-}{1-t^-} \frac{\delta}{(1-\alpha)^2\bar{K}} \left\{ \frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} - \log \left(\frac{\tilde{A} - r/\alpha}{A_l - r/\alpha} \right) - 1 \right\}. \quad (31)$$

Because $A'(s, t^+, t^-) \leq x^*$,

$$A'(s, t^+, t^-) = \min\{\tilde{A}(s, t^+, t^-), x^*\}.$$

For the upper cutoff x'' , for a small noise, $x'' = \bar{A} + \sigma$ for $s/t^+ \geq \bar{A} - A_l = \pi(\bar{K}, \bar{A})$ because $t^+\pi(\bar{K}, \bar{A}) - s = \pi_{gpls}(2, \bar{K}, \bar{A}) - \pi_{gpls}(1, \bar{K}, \bar{A}) \leq 0$. Therefore, as the noise becomes vanishingly small, $x'' = \bar{A}$.

Finally, when $s/t^+ < \bar{A} - A_l = \pi(\bar{K}, \bar{A})$, from Proposition 11, $x'' = \max\{x', x^\diamond\}$ where $x^\diamond$ is the minimum x that solves $s = E[(t^+ \max\{\pi(K(A; x'), A), 0\} + t^- \min\{\pi(K(A; x'), A), 0\}) | x_i = x]$. Because $t^- \min\{\pi(K(A; x'), A), 0\} \leq 0$, $x^\diamond$ satisfies $s \leq t^+ E[\max\{\pi(K(A; x'), A), 0\} | x_i = x^\diamond]$. Then, consider the two cases. First, suppose $A'(s, t^+, t^-) < A_l + \frac{s}{t^+}$. In this case $s/t^+ > 0$. Then, for sufficiently small σ , $x'(s, t^+, t^-) + 3\sigma < A_l + \frac{s}{t^+}$. It follows that $\frac{s}{t^+} \leq E[\max\{\pi(K(A; x'), A), 0\} | x_i = x^\diamond] \leq E[\max\{A - A_l, 0\} | x_i = x^\diamond]$. Also, in this case,

because $s/t^+ > 0$, $x^\diamond \geq A_l - \sigma$. Then, $E[\max\{A - A_l, 0\}|x_i = x^\diamond] \leq x^\diamond - (A_l - \sigma)$. Therefore, $\frac{s}{t^+} \leq E[\max\{\pi(K(A; x'), A), 0\}|x_i = x^\diamond] \leq x^\diamond - (A_l - \sigma) \implies x^\diamond \geq A_l + \frac{s}{t^+} - \sigma$. Then, using the same arguments as in the proof of [Proposition 10](#), $x''(s, t^+, t^-) > x'(s, t^+, t^-) + 2\sigma$. Additionally, for $\sigma < \frac{s}{2t^+}$, $x'' = x^\diamond > A_l + \sigma$. Hence, $E[\min\{\pi(K(A; x'), A), 0\}|x_i = x''] = 0$ and $E[\max\{\pi(K(A; x'), A), 0\}|x_i = x''] = E[\pi(K(A; x'), A)|x_i = x'']$. Thus, the upper cutoff x'' solves

$$s = t^+ E[\max\{\pi(K(A; x'), A), 0\}|x_i = x''] = t^+ E[\pi(K(A; x'), A)|x_i = x''].$$

Therefore, in this case, the upper cutoff is $x''(s, t^+, t^-) = A_l + \frac{s}{t^+}$, and as the noise becomes vanishingly small,

$$\lim_{\sigma \rightarrow 0} x''(s, t^+, t^-) = A''(s, t^+, t^-) = \frac{r}{\alpha} + \frac{\delta}{(1 - \alpha)^2} \frac{1}{\underline{K} + \overline{K}} + \frac{s}{t^+}. \quad (32)$$

Next, consider the case when $A'(s, t^+, t^-) > A_l + \frac{s}{t^+}$. Using the same arguments as in the proof of [Proposition 10](#), we can show that

$$A''(s, t^+, t^-) = A'(s, t^+, t^-). \quad (33)$$

Finally, for $A'(s, t^+, t^-) = A_l + \frac{s}{t^+}$, the two candidate limits coincide. Thus, as in the GPS program, for any GPLS program (s, t^+, t^-) , as the noise vanishes, $A''(s, t^+, t^-)$ is given by either equation (32) or equation (33). By the same arguments used to compute A'' in the GPS case, we obtain

$$A''(s, t^+, t^-) = \begin{cases} A_l + s/t^+ & ; s \geq t^+ (A'(s, t^+, t^-) - A_l) \\ A'(s, t^+, t^-) & ; s \leq t^+ (A'(s, t^+, t^-) - A_l). \end{cases}$$

This concludes the proof. □

Proof of [Proposition 8](#). We prove this in three parts:

1. Show that for a given t^+ and t^- , there is a unique $s(x', t^+, t^-)$ that induces x' .
2. Show that the expected cost of implementing any target x' is decreasing in both t^+ and t^- .
3. Compute the expected cost of resolving miscoordination with the optimal GPLS program as the noise becomes vanishingly small.

Lemma 6. Under assumptions 1 and 4, for any $x' \in (x^* - \frac{\bar{s}}{1-\bar{t}}, x^*)$, the set of GPLS programs $\{(s, t^+, t^-) : s \in [0, \bar{s}], t^+ \in [0, 1], t^- \in [0, \bar{t}]\}$ that implement the target x' is non-empty. Furthermore, for a given t^+ and t^- , if x' is implementable by that t^+ and t^- , there is a unique s , denoted by $s(x', t^+, t^-)$, that implements x' .

Proof. The proof for this follows the same approach as the proof for lemma 3 with some modifications. Any $x' \in (x^* - \frac{\bar{s}}{1-\bar{t}}, x^*)$ will either satisfy $x' < A_0(x', \sigma) - \sigma$ or $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$. When $x' < A_0(x', \sigma) - \sigma$, $A_0(x', \sigma) = A_l$ (from equation (15)). It follows that $x' < A_l - \sigma$. Then, from Proposition 11, x' is $x' = x'(s, t^+, t^-) = x^* - \frac{s}{1-t^-}$. For any $x' < A_0(x', \sigma) - \sigma$, the set of GPLS program that induces x' is $\{(s, t^+, t^-) : s = (1 - t^-)(x^* - x'), t^+ \in [0, 1], t^- \in [0, \bar{t}]\}$, and for any t^+ and t^- , the guarantee s that induces x' is,

$$s(x', t^+, t^-) = (1 - t^-)(x^* - x'). \quad (34)$$

When $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$, for any $x' < x^*$, any (s, t^+, t^-) that satisfies $x' - T_2(x') = 0$ implements x' (Proposition 11), where

$$\begin{aligned} T_2(x') &= \frac{t^+}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \frac{x' - A_0(x', \sigma)}{\sigma}\right)} \right) \right] \\ &\quad + \frac{t^-}{2\sigma} \left[\frac{A_0(x', \sigma)^2 - (x' - \sigma)^2}{2} - \frac{r(A_0(x', \sigma) - x' + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \frac{\bar{K}}{2} \left(1 - \frac{x' - A_0(x', \sigma)}{\sigma}\right)}{\underline{K}} \right) \right] \\ &\quad + x^* - s \\ &= \frac{\gamma}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \frac{x' - A_0(x', \sigma)}{\sigma}\right)} \right) \right] \\ &\quad + t^-(x' - x^*) + x^* - s. \end{aligned}$$

The derivative of $T_2(x')$ w.r.t. x' is $T_2'(x') = \frac{\gamma}{2\sigma}(x' + \sigma - A_0(x', \sigma)) + t^-$. Because $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$, $(x' + \sigma - A_0(x', \sigma)) \in [0, 2\sigma]$. Substituting $\gamma = t^+ - t^-$, we have $T_2'(x') \in [0, 1]$. Then, by the implicit function formula, $\frac{dx'}{ds} = -\frac{1}{1-T_2'(x')} < 0$, which implies that for a given t^+ and t^- , there is a one-to-one mapping from x' to s . Therefore, there exists a set of programs (s, t^+, t^-) that induce any $x' \in (x^* - \frac{\bar{s}}{1-\bar{t}}, x^*)$ and for a given t^+ and t^- , there is a unique $s(x', t^+, t^-)$ that induces x' . Furthermore, when $x' = A_0(x', \sigma) - \sigma$, the unique $s(x', t^+, t^-)$ that induces x' is given by equation (34). $\square$

The following lemma proves the second part:

Lemma 7. *Under assumptions 1 and 4, the expected cost to the policymaker to implement a target x' , where $x' \in (x^* - \frac{\bar{s}}{1-t^-}, x^*)$, is (weakly) decreasing in both t^+ and t^- .*

Proof. By lemma 6, we can state the GPLS program (s, t^+, t^-) in terms of x' , t^+ , and t^- . Let $C(A, x', t^+, t^-; \sigma)$ denote the cost of inducing x' for given t^+ , t^- , and A . Using the same approach as in lemma 4, the expected cost of the program (x', t^+, t^-) is

$$\begin{aligned} E_A[C(A, x', t^+, t^-; \sigma)] &= \int_{x'}^{x''} \int_{\underline{A}}^{\bar{A}} [\pi_{gpls}(1, K(A; x'), A) - \pi_{gpls}(2, K(A; x'), A)] f(A|x_i) dA f(x_i) dx_i \\ &= \int_{x'}^{x''} (U_1^{gpls}(x', x_i) - U_2^{gpls}(x', x_i)) f(x_i) dx_i \\ &= \int_{x'}^{\bar{A}+\sigma} \max\{U_1^{gpls}(x', x_i) - U_2^{gpls}(x', x_i), 0\} f(x_i) dx_i, \end{aligned} \quad (35)$$

where

$$U_1^{gpls}(x', x_i) = \int_{\underline{A}}^{\bar{A}} \pi_{gpls}(1, K(A; x'), A) f(A | x_i) dA, \quad U_2^{gpls}(x', x_i) = \int_{\underline{A}}^{\bar{A}} \pi_{gpls}(2, K(A; x'), A) f(A | x_i) dA.$$

Because U_2^{gpls} does not depend on t^+ or t^- , the relationship between the cost of implementing x' and t^+, t^- depends on how U_1^{gpls} varies with t^+ and t^- for $x_i > x'$. For a given x' ,

$$U_1^{gpls}(x', x_i) = U_2^{gpls}(x', x_i) - \left(\int_{\underline{A}}^{\bar{A}} t^+ \max\{\pi(K(A, x'), A), 0\} + t^- \min\{\pi(K(A, x'), A), 0\} f(A|x_i) dA \right) + s.$$

Since U_2^{gpls} and π are independent of t^+ and t^- ,

$$\begin{aligned} \frac{d}{dt^+} U_1^{gpls}(x', x_i) &= - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A, x'), A), 0\} f(A|x_i) dA + \frac{ds(x', t^+, t^-)}{dt^+}, \\ \frac{d}{dt^-} U_1^{gpls}(x', x_i) &= - \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A, x'), A), 0\} f(A|x_i) dA + \frac{ds(x', t^+, t^-)}{dt^-}. \end{aligned}$$

Next, consider the two cases:

Case 1: $x' < A_0(x', \sigma) - \sigma$

From equation (34), $s(x', t^+, t^-) = (1 - t^-)(x^* - x')$. Then, for a fixed x' , $\frac{ds(x', t^+, t^-)}{dt^-} = -(x^* - x') = x' - x^*$. Then,

$$\frac{d}{dt^-} U_1^{gpls}(x', x) = x' - x^* - \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A, x'), A), 0\} f(A|x_i) dA.$$

Since $\int_{\underline{A}}^{\bar{A}} \pi(K(A, x'), A) f(A|x_i) dA$, the expected payoff with x' as the cutoff strategy, is increasing in x_i then for any $x \geq x'$,

$$\begin{aligned} \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A, x'), A), 0\} f(A|x_i) dA &\geq \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A, x'), A), 0\} f(A|x') dA \\ &\geq \int_{\underline{A}}^{\bar{A}} \pi(K(A, x'), A) f(A|x') dA \quad \because \{x' < A_0(x', \sigma) - \sigma\} \\ &\geq x' - x^*. \end{aligned}$$

This implies that $\frac{d}{dt^-} U_a^{gpls}(x', x_i) = x' - x^* - \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A, x'), A), 0\} f(A|x_i) dA \leq 0$ for all $x_i > x'$. So, the least costly way to implement $x' < A_0(x', \sigma) - \sigma$ is with the highest t^- .

Next, for a fixed x' , $\frac{ds(x', t^+, t^-)}{dt^+} = 0$ and $\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A, x'), A), 0\} f(A|x_i) dA \geq 0$. Then, $\frac{d}{dt^+} U_1^{gpls}(x', x_i) \leq 0$ for all $x > x'$. So, the least costly way to implement $x' < A_0(x', \sigma) - \sigma$ is with the highest t^+ .

Case 2: $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$

For any given x' , when investors receive that signal, their expected payoff must be 0. Then,

$$s = t^+ \cdot E[\max\{\pi(K(A; x'), A), 0\}|x'] + t^- \cdot E[\min\{\pi(K(A; x'), A), 0\}|x'] - E[\pi(K(A; x'), A)|x'].$$

Taking derivatives w.r.t. t^+ & t^- ,

$$\begin{aligned} \frac{ds(x', t^+, t^-)}{dt^+} &= E[\max\{\pi(K(A; x'), A), 0\}|x'] = \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x') dA, \\ \frac{ds(x', t^+, t^-)}{dt^-} &= E[\min\{\pi(K(A; x'), A), 0\}|x'] = \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x') dA. \end{aligned}$$

First consider t^+ . In this case,

$$\begin{aligned} \frac{d}{dt^+} U_a^{gpls}(x', x_i) &= - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + \frac{ds(x', t^+, t^-)}{dt^+} \\ &= - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x') dA. \end{aligned}$$

Now, $\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA$ is weakly increasing in x_i . Furthermore, we only need to check for $x_i > x'$ since that is the range over which we integrate in equation (35).

Then, for any $x_i > x'$, $\int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA \geq \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x') dA$.

This implies that

$$\frac{d}{dt^+} U_1^{gpls}(x', x_i) = - \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + \int_{\underline{A}}^{\bar{A}} \max\{\pi(K(A; x'), A), 0\} f(A|x') dA \leq 0.$$

Therefore, the cost of implementing any $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$ is decreasing in t^+ . Finally, consider t^- .

$$\begin{aligned} \frac{d}{dt^-} U_1^{gpls}(x', x_i) &= - \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + \frac{ds(x', t^+, t^-)}{dt^-} \\ &= - \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x_i) dA + \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x') dA. \end{aligned}$$

The term $\int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x_i) dA$ is weakly increasing in x_i . Therefore, for any $x_i > x'$, $\int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x_i) dA \geq \int_{\underline{A}}^{\bar{A}} \min\{\pi(K(A; x'), A), 0\} f(A|x') dA$. This implies that $\frac{d}{dt^-} U_1^{gpls}(x', x_i) \leq 0$.

Therefore, the cheapest way to implement any target x' is to set the highest possible value for t^+ & t^- . $\square$

Finally, we compute the expected cost of removing miscoordination. The guarantor must set a target $x' = A_l - \sigma$. Assumptions 4 and 3 ensure that it is feasible to achieve this target, i.e., $x^* - A_l + \sigma \leq \frac{\bar{s}}{1-\bar{t}} < A_l - \underline{A} - \sigma$. Because the cost is decreasing in both t^+ and t^- , she can set $t^+ = 1$ and $t^- = \bar{t}$. The corresponding guarantee is $s(x', 1, \bar{t}) = (1 - \bar{t})(x^* - A_l + \sigma)$ (from Proposition 11). For a small enough σ , $\sigma < (1 - \bar{t})(x^* - A_l + \sigma)$, the upper cutoff is $x'' > A_l + \sigma$ (computing the expectation in Proposition 11). Then, for $x' = A_l - \sigma$, the expected cost is

$$\begin{aligned}
E_A[C(A, x', 1, \bar{t}; \sigma)] &= \int_{\underline{A}}^{\bar{A}} \left((s(x', 1, \bar{t}) - \max\{\pi(K(A; x'), A), 0\}) \right. \\
&\quad \left. - \bar{t} \cdot \min\{\pi(K(A; x'), A), 0\}) \cdot (F(x''|A) - F(x'|A)) \right) f(A) dA \\
&= \frac{1}{\bar{A} - \underline{A}} \left\{ 0 + \int_{x'-\sigma}^{A_0=x'+\sigma} (s(x', 1, \bar{t}) - \bar{t} \cdot \pi(K(A; x'), A)) \left(\frac{1}{2} - \left(\frac{x' - A}{2\sigma} \right) \right) dA \right. \\
&\quad + \int_{A_0=x'+\sigma}^{x''-\sigma} (s(x', 1, \bar{t}) - \pi(K(A; x'), A)) dA \\
&\quad \left. + \int_{x''-\sigma}^{x''+\sigma} (s(x', 1, \bar{t}) - \pi(K(A; x'), A)) \left(\frac{1}{2} \left(\frac{x'' - A}{\sigma} \right) + \frac{1}{2} \right) dA + 0 \right\}.
\end{aligned}$$

We compute each of the three terms individually. Consider the first term,

$$\begin{aligned}
&\int_{x'-\sigma}^{x'+\sigma} (s(x', 1, \bar{t}) - \bar{t} \cdot \pi(K(A; x'), A)) \left(\frac{1}{2} - \left(\frac{x' - A}{2\sigma} \right) \right) dA \\
&= \int_{x'-\sigma}^{x'+\sigma} \left[s(x', 1, \bar{t}) - \bar{t} \left(A - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2(\underline{K} + \frac{\bar{K}}{2}(1 - (\frac{x'-A}{\sigma}))} \right) \right] \left(\frac{1}{2} - \left(\frac{x' - A}{2\sigma} \right) \right) dA \\
&= \sigma \left[s(x', 1, \bar{t}) - \bar{t} \left[x' - \frac{r}{\alpha} + \frac{\sigma}{3} - \frac{2\delta}{(1-\alpha)^2\bar{K}} + \frac{2\delta\underline{K}}{(1-\alpha)^2\bar{K}^2} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) \right] \right].
\end{aligned}$$

For the second term,

$$\begin{aligned}
\int_{x'+\sigma}^{x''-\sigma} (s(x', 1, \bar{t}) - \pi(K(A; x'), A)) dA &= \int_{x'+\sigma}^{x''-\sigma} \left(s(x', 1, \bar{t}) - \left(A - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2(\underline{K} + \bar{K})} \right) \right) dA \\
&= (x'' - x' - 2\sigma) \left[s(x', 1, \bar{t}) - \left(\frac{x'' + x'}{2} - A_l \right) \right].
\end{aligned}$$

Finally, consider the third term,

$$\begin{aligned}
&\int_{x''-\sigma}^{x''+\sigma} (s(x', 1, \bar{t}) - \pi(K(A; x'), A)) \left(\frac{1}{2} \left(\frac{x'' - A}{\sigma} \right) + \frac{1}{2} \right) dA \\
&= \int_{x''-\sigma}^{x''+\sigma} \left(s(x', 1, \bar{t}) - \left(A - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2(\underline{K} + \bar{K})} \right) \right) \left(\frac{1}{2} \left(\frac{x'' - A}{\sigma} \right) + \frac{1}{2} \right) dA \\
&= \sigma \left(s(x', 1, \bar{t}) - \left(x'' - A_l - \frac{\sigma}{3} \right) \right).
\end{aligned}$$

Then, the expected cost is

$$\begin{aligned}
E_A[C(A, x', 1, \bar{t}; \sigma)] &= \frac{1}{\bar{A} - \underline{A}} \left\{ \sigma \left[s(x', 1, \bar{t}) - \bar{t} \left[x' - \frac{r}{\alpha} + \frac{\sigma}{3} - \frac{2\delta}{(1-\alpha)^2\bar{K}} + \frac{2\delta\underline{K}}{(1-\alpha)^2\bar{K}^2} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) \right] \right] \right. \\
&\quad \left. + (x'' - x' - 2\sigma) \left[s(x', 1, \bar{t}) - \left[\frac{x'' + x'}{2} - A_l \right] \right] + \sigma \left[s(x', 1, \bar{t}) - \left(x'' - A_l - \frac{\sigma}{3} \right) \right] \right\}.
\end{aligned}$$

where $s(x', 1, \bar{t}) = (1 - \bar{t})(x^* - A_l + \sigma)$. As the noise goes to 0, the cost is

$$\lim_{\sigma \rightarrow 0} E_A[C(A, x', 1, \bar{t}; \sigma)] = \lim_{\sigma \rightarrow 0} \frac{(x'' - x')}{\bar{A} - \underline{A}} \left[s(x', 1, \bar{t}) - \left(\frac{x'' + x'}{2} - A_l \right) \right].$$

As the noise becomes vanishingly small, $x' = A' = A_l, x'' = A'' = A_l + (1 - \bar{t})(x^* - A_l)$ (from [Proposition 7](#)), and $\lim_{\sigma \rightarrow 0} s(x', 1, \bar{t}) = (1 - \bar{t})(x^* - A_l)$. The expected cost of removing miscoordination is

$$\begin{aligned} \lim_{\sigma \rightarrow 0} E_A[C(A, x', 1, \bar{t}; \sigma)] &= \frac{(A'' - A')}{\bar{A} - \underline{A}} \left[(1 - \bar{t})(x^* - A_l) - \left(\frac{A'' + A'}{2} - A_l \right) \right] \\ &= \frac{(1 - \bar{t})(x^* - A_l)}{\bar{A} - \underline{A}} \left[\frac{(1 - \bar{t})(x^* - A_l)}{2} \right] = \frac{(1 - \bar{t})^2 (x^* - A_l)^2}{2(\bar{A} - \underline{A})}. \end{aligned} \quad (36)$$

Therefore, as the noise becomes vanishingly small, the optimal GPLS program that removes miscoordination is $(s, t^+, t^-) = ((1 - \bar{t})(x^* - A_l), 1, \bar{t})$, and its expected cost is $\frac{(1 - \bar{t})^2 (x^* - A_l)^2}{2(\bar{A} - \underline{A})}$. $\square$

Proof of [Proposition 9](#). The R-GPLS program is a GPLS program with $t^+ = t^-$. Thus, lemma [7](#) holds for the R-GPLS program which implies that the optimal R-GPLS program sets $t^+ = t^- = \bar{t}$. By the same argument provided in the proof of [Proposition 8](#), the guarantee s required to remove miscoordination ($x' = A_l - \sigma$) is,

$$s(A_l - \sigma, \bar{t}, \bar{t}) = (1 - \bar{t})(x^* - A_l + \sigma). \quad (37)$$

The expected cost calculation is identical to the optimal GPLS case, except that here we set $t^+ = \bar{t}$ (instead of $t^+ = 1$). Then, as the noise becomes vanishingly small, $x' = A' = A_l$ and $\lim_{\sigma \rightarrow 0} s(x', \bar{t}, \bar{t}) = (1 - \bar{t})(x^* - A_l)$. Consider the following two cases:

Case 1: $\bar{t} > \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$

In this case, $\bar{t} > \frac{x^* - A_l}{\bar{A} + x^* - 2A_l} \implies (1 - \bar{t})(x^* - A_l) < \bar{t}(\bar{A} - A_l)$. Thus, $\lim_{\sigma \rightarrow 0} s(x', \bar{t}, \bar{t}) < \bar{t}(\bar{A} - A_l)$. It follows from [Proposition 7](#) that $A'' < \bar{A}$. By the same proposition, because $\lim_{\sigma \rightarrow 0} s(x', \bar{t}, \bar{t}) > 0$, the upper cutoff, as the noise becomes vanishingly small, is $x'' = A'' = A_l + \frac{(1 - \bar{t})}{\bar{t}}(x^* - A_l)$. Using similar calculations as in the proof of [Proposition 8](#), the expected cost of removing miscoordination, as the noise becomes vanishingly small, is

$$\begin{aligned}\lim_{\sigma \rightarrow 0} E_A[C(A, x', \bar{t}, \bar{t}; \sigma)] &= \frac{(A'' - A')}{\bar{A} - \underline{A}} \left[(1 - \bar{t})(x^* - A_l) - \bar{t} \left(\frac{A'' + A'}{2} - A_l \right) \right] \\ &= \frac{(1 - \bar{t})}{\bar{t}} \frac{(x^* - A_l)}{\bar{A} - \underline{A}} \left[\frac{(1 - \bar{t})(x^* - A_l)}{2} \right] = \frac{(1 - \bar{t})^2}{2\bar{t}} \frac{(x^* - A_l)^2}{\bar{A} - \underline{A}}.\end{aligned}$$

For $\frac{x^* - A_l}{\bar{A} + x^* - 2A_l} < \bar{t} < 1$, $\frac{(1 - \bar{t})^2}{2\bar{t}} \frac{(x^* - A_l)^2}{\bar{A} - \underline{A}} > \frac{(1 - \bar{t})^2}{2} \frac{(x^* - A_l)^2}{\bar{A} - \underline{A}}$

Case 2: $\bar{t} \leq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$

In this case, $\bar{t} \leq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l} \implies (1 - \bar{t})(x^* - A_l) \geq \bar{t}(\bar{A} - A_l)$. That is, $\lim_{\sigma \rightarrow 0} s(x', \bar{t}, \bar{t}) \geq \bar{t}(\bar{A} - A_l)$.

Therefore, $A'' = \bar{A}$ (from [Proposition 7](#)). Using similar calculations as in the proof of [Proposition 8](#), the expected cost, as the noise becomes vanishingly small, is

$$\begin{aligned}\lim_{\sigma \rightarrow 0} E_A[C(A, x', \bar{t}, \bar{t}; \sigma)] &= \frac{(A'' - A')}{\bar{A} - \underline{A}} \left[(1 - \bar{t})(x^* - A_l) - \bar{t} \left(\frac{A'' + A'}{2} - A_l \right) \right] \\ &= \left(\frac{\bar{A} - A_l}{\bar{A} - \underline{A}} \right) \left[(1 - \bar{t})(x^* - A_l) - \frac{\bar{t}}{2}(\bar{A} - A_l) \right]\end{aligned}$$

Finally, we check if the expected cost in this case is greater than the expected cost for the optimal GPLS program. The difference in the expected costs between the optimal GPLS and the optimal R-GPLS is

$$D(\bar{t}) = \frac{(1 - \bar{t})^2}{2} \frac{(x^* - A_l)^2}{\bar{A} - \underline{A}} - \left(\frac{\bar{A} - A_l}{\bar{A} - \underline{A}} \right) \left[(1 - \bar{t})(x^* - A_l) - \frac{\bar{t}}{2}(\bar{A} - A_l) \right].$$

$D(\bar{t})$ is quadratic and convex in $\bar{t}$ (the second derivative with respect to $\bar{t}$ is $\frac{(x^* - A_l)^2}{\bar{A} - \underline{A}} > 0$).

Therefore, for $\bar{t} \in [0, \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}]$, it suffices to check if $D(\bar{t}) < 0$ at $\bar{t} = 0$ and $\bar{t} = \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$.

First, we check at $\bar{t} = 0$. The difference is $D(0) = \left(\frac{x^* - A_l}{\bar{A} - \underline{A}} \right) \left[\frac{x^* - A_l}{2} - (\bar{A} - A_l) \right] < 0$ because $(x^* - A_l) < 2(\bar{A} - A_l)$.

For $\bar{t} = \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$, $D(\bar{t}) = \left(\frac{\bar{A} - A_l}{\bar{A} - A_l + x^* - A_l} \right)^2 \cdot \frac{(x^* - A_l)^2}{2(\bar{A} - \underline{A})} - \frac{(\bar{A} - A_l)}{2(\bar{A} - \underline{A})} \left[\frac{2(\bar{A} - A_l)(x^* - A_l) - (x^* - A_l)(\bar{A} - A_l)}{\bar{A} - A_l + x^* - A_l} \right]$. We

can rewrite the difference as $D(\bar{t}) = \frac{(\bar{A} - A_l)^2}{2(\bar{A} - \underline{A})} \cdot \frac{x^* - A_l}{\bar{A} - A_l + x^* - A_l} \left[\frac{x^* - A_l}{\bar{A} - A_l + x^* - A_l} - 1 \right] < 0$ because

$(x^* - A_l) - (\bar{A} - A_l + x^* - A_l) = A_l - \bar{A} < 0$. Therefore, for any $\bar{t} \leq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$, $D(\bar{t}) < 0$ and,

thus, $\left(\frac{\bar{A} - A_l}{\bar{A} - \underline{A}} \right) \left[(1 - \bar{t})(x^* - A_l) - \frac{\bar{t}}{2}(\bar{A} - A_l) \right] > \frac{(1 - \bar{t})^2}{2} \frac{(x^* - A_l)^2}{\bar{A} - \underline{A}}$.

At $\bar{t} = \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$, $\frac{(1-\bar{t})^2}{2\bar{t}} \frac{(x^* - A_l)^2}{\bar{A} - A_l} = \frac{\bar{A} - A_l}{\bar{A} - A_l} \cdot \frac{(\bar{A} - A_l)(x^* - A_l)}{2(\bar{A} + x^* - 2A_l)} = \left(\frac{\bar{A} - A_l}{\bar{A} - A_l} \right) \left[(1 - \bar{t})(x^* - A_l) - \frac{\bar{t}}{2}(\bar{A} - A_l) \right]$.
Thus, as $\sigma \rightarrow 0$, the optimal R-GPLS program that removes miscoordination is $(s, t^-, t^-) = ((1 - \bar{t})(x^* - A_l), \bar{t}, \bar{t})$, and its expected cost is

$$\lim_{\sigma \rightarrow 0} E_A[C(A, x', \bar{t}, \bar{t}; \sigma)] = \begin{cases} \frac{1}{\bar{t}} \frac{(1-\bar{t})^2}{2} \frac{(x^* - A_l)^2}{\bar{A} - A_l} & ; \bar{t} \geq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l} \\ \left(\frac{\bar{A} - A_l}{\bar{A} - A_l} \right) \left[(1 - \bar{t})(x^* - A_l) - \frac{\bar{t}}{2}(\bar{A} - A_l) \right] & ; \bar{t} \leq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}, \end{cases} \quad (38)$$

which is strictly larger than the expected cost for the optimal GPLS program when $\bar{t} < 1$. $\square$

Proof of Corollary 3. It follows directly from Proposition 8 and Proposition 5 that $\frac{C(GPLS)}{C(GPS)} = (1 - \bar{t})^2$. For comparison between $C(GPLS)$ and $C(SZ)$, denote $\frac{C(GPLS)}{C(SZ)} = m(\bar{t})$. When $\bar{t} \geq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$, it follows from Proposition 8 and Proposition 9 that $m(\bar{t}) = \bar{t}$. Therefore, $m(\bar{t}) \leq \max\{\bar{t}, \frac{x^* - A_l}{\bar{A} - A_l}\}$.

When $\bar{t} \leq \frac{x^* - A_l}{\bar{A} + x^* - 2A_l}$, from equations (36) and (38),

$$\begin{aligned} m(\bar{t}) &= \frac{(1 - \bar{t})^2 (x^* - A_l)^2}{2(\bar{A} - A_l) \left[(1 - \bar{t})(x^* - A_l) - \frac{\bar{t}}{2}(\bar{A} - A_l) \right]} = \left(\frac{x^* - A_l}{\bar{A} - A_l} \right)^2 \frac{1 - \bar{t}}{2 \left(\frac{x^* - A_l}{\bar{A} - A_l} \right) - \frac{\bar{t}}{1 - \bar{t}}} \\ &< \left(\frac{x^* - A_l}{\bar{A} - A_l} \right)^2 \frac{1}{2 \left(\frac{x^* - A_l}{\bar{A} - A_l} \right) - \frac{x^* - A_l}{\bar{A} - A_l}} = \frac{x^* - A_l}{\bar{A} - A_l}, \end{aligned}$$

where the inequality uses $\frac{\bar{t}}{1 - \bar{t}} < \frac{x^* - A_l}{\bar{A} - A_l}$. $\square$

Proof of Corollary 4. Under optimal GPS and GPLS programs, (1) nearly all invest when $A > A_l$ and nearly none do when $A < A_l$, and (2) the normalized net payoff from investing versus not investing is $\pi(\underline{K}, A) < 0$ for $A < A_l$, and $\pi(\bar{K}, A) > 0$ for $A > A_l$. Therefore, from (8) and (3), a participating investor's return under the optimal GPS program is

$$r_{gps} = \alpha \pi(1, K, A) + r = \begin{cases} \alpha s_{gps} + \alpha \left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K}} \right) & ; A < A_l \\ \alpha s_{gps} + r & ; A > A_l \end{cases} \quad (39)$$

$$= \begin{cases} r - \alpha (A_h - x^*) - \alpha (A_l - A) & ; A < A_l \\ r + \alpha (x^* - A_l) & ; A > A_l, \end{cases} \quad (40)$$

where $s_{gps} = x^* - A_l$ (Proposition 5). For $A < A_l$, we have $r_{gps} < r$, because $x^* \leq A_h$.

Similarly, from (14) and (3), the corresponding return under the optimal GPLS program is

$$r_{gpls} = \alpha \pi_{gpls}(1, K, A) + r = \begin{cases} \alpha s_{gpls} + \bar{t} r + \alpha(1 - \bar{t}) \left(A - \frac{\delta}{(1-\alpha)^2 \underline{K}} \right) & ; A < A_l \\ \alpha s_{gpls} + r & ; A > A_l \end{cases} \quad (41)$$

$$= \begin{cases} (1 - \bar{t}) r_{gps} + \bar{t} r & ; A < A_l \\ r + (1 - \bar{t}) \alpha (x^* - A_l) & ; A > A_l, \end{cases} \quad (42)$$

where the second line is strictly larger than the first line since $r > r_{gps}$ when $A < A_l$ (the second line can be written as $(1 - \bar{t}) r_{gps} + \bar{t} r + (1 - \bar{t})(r - r_{gps} + \alpha (x^* - A_l))$). Here we used $s_{gpls} = (1 - \bar{t}) s_{gps}$ since $s_{gpls} = (1 - \bar{t})(x^* - A_l)$ from Proposition 8.

The returns for investors who do not participate is similarly

$$r_{np} = \alpha \pi(K, A) + r = \begin{cases} \alpha \left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K}} \right) & ; A < A_l \\ \alpha \left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \bar{K}} \right) & ; A > A_l. \end{cases} \quad (43)$$

$$= \begin{cases} r - \alpha (A_h - A) & ; A < A_l \\ r + \alpha (A - A_l) & ; A > A_l. \end{cases} \quad (44)$$

□

Proof of Proposition 10. Fix (s, t) . Throughout this proof, to simplify notation, we use x' and x'' to denote the cutoffs induced by (s, t) . Then, given that action and state monotonicity hold (see the analysis preceding Proposition 4), the following lemma follows from standard arguments (Frankel, Morris, and Pauzner, 2003, Van Zandt and Vives, 2007).

Lemma 8. *Player i 's best response to a monotone strategy profile is a monotone strategy.*

The following lemma formally defines a symmetric monotone equilibrium in terms of a cutoff pair.

Lemma 9. *A cutoff pair (x', x'') constitutes a symmetric monotone equilibrium if and only if*

$$x' = \min \{ \min \{ x \text{ s.t. } \Delta^1(x; x') \geq 0 \}, x^* \} \quad (45)$$

$$x'' = \max \{ \min \{ \{ x \text{ s.t. } \Delta^2(x; x') \geq 0 \} \cup \{ \bar{A} + \sigma \} \}, x' \},$$

where $\Delta^1(x; x') = \mathbb{E}[\delta^1(K(A; x'), A) | x_i = x]$, $\Delta^2(x; x') = \mathbb{E}[\delta^2(K(A; x'), A) | x_i = x]$ and x^* is the benchmark cutoff defined in [Proposition 2](#).

Proof of Lemma 9. Consider investor i with signal x_i and fix other investors' cutoff strategy as (x', x'') . From [Lemma 8](#), player i 's best response to a monotone strategy (x', x'') will be a monotone strategy characterized by two cutoffs. Denote i 's best response to (x', x'') by a cutoff pair $(x'_i(x', x''), x''_i(x', x''))$ with $x'_i(x', x'') \leq x''_i(x', x'')$, such that investor i chooses action 0 if $x_i < x'_i(x', x'')$, chooses action 1 if $x_i \in [x'_i(x', x''), x''_i(x', x'')]$, and chooses action 2 if $x_i \geq x''_i(x', x'')$, where we maintain our tie-breaking rule that the investor chooses action 1 when indifferent between actions 0 and 1, and chooses 2 when indifferent between actions 2 and any other action. A pair (x', x'') constitutes a symmetric monotone equilibrium if and only if

$$x' = x'_i(x', x'') \quad \text{and} \quad x'' = x''_i(x', x''). \quad (46)$$

From equation (9), investor i 's relative payoffs from choosing action 1 over 0 and 2 over 1, respectively, are

$$\delta^1(K(A; x'), A) = \pi(K(A; x'), A) - t \cdot \max\{\pi(K(A; x'), A), 0\} + s$$

$$\delta^2(K(A; x'), A) = t \cdot \max\{\pi(K(A; x'), A), 0\} - s.$$

Her relative payoff from choosing action 2 over 0 is

$$\delta^{bench}(K(A; x'), A) = \pi(2, K(A; x'), A) = \pi(K(A; x'), A).$$

Suppose investor i receives a signal $x_i = x$, then her expected relative payoffs are

$$\Delta^1(x; x') = \mathbb{E}[\delta^1(K(A; x'), A) | x_i = x] \quad ; \quad \Delta^2(x; x') = \mathbb{E}[\delta^2(K(A; x'), A) | x_i = x]$$

$$\Delta^{bench}(x; x') = \mathbb{E}[\delta^{bench}(K(A; x'), A) | x_i = x] = \mathbb{E}[\pi(K(A; x'), A) | x_i = x].$$

For $x \in [\underline{A} - \sigma, \bar{A} + \sigma)$, investor i chooses action 2 over action 1 if and only if $\Delta^2(x; x') \geq 0$, she chooses action 1 over action 0 if and only if $\Delta^1(x; x') \geq 0$, and chooses action 2 over action 0 if and only if $\Delta^{bench}(x; x') \geq 0$.

First, consider investor i 's decision about choosing action 1 over 0. The relative payoff from choosing action 1 over 0, $\delta^1(K(A; x'), A)$, is weakly increasing in A because $t \leq 1$ and $\pi(K(A; x'), A)$ is strictly increasing in A . Because $F(A|x)$ is increasing with x in the FOSD sense, $\Delta^1(x; x')$ is weakly increasing in x . Moreover, because $\delta^1(K(A; x'), A)$ is bounded in A (it is a piecewise linear function of $\pi(K(A; x'), x')$ which is bounded in A) and $F(A|x)$ is continuous in x , $\Delta^1(x; x')$ is continuous in x . Then, by [Assumption 2](#) and [Assumption 1.3](#), there exists a minimum x that satisfies $\Delta^1(x; x') \geq 0$. Denote it by $x_i^l(x')$ so that

$$x_i^l(x') = \min\{x \text{ s.t. } \Delta^1(x; x') \geq 0\}. \quad (47)$$

Then, $\Delta^1(x; x') \geq 0$ if and only if $x \geq x_i^l(x')$, and therefore, player i chooses action 1 over 0 if and only if $x \geq x_i^l(x')$.

Second, consider i 's relative payoff from choosing action 2 over 0. $\Delta^{bench}(x; x')$ is strictly increasing in x because $\pi(K(A; x'), A)$ is strictly increasing in A and $F(A|x)$ is increasing in x in the FOSD sense. Additionally, because $\pi(K(A; x'), A)$ is bounded in A and $F(A|x)$ is continuous in x , $\Delta^{bench}(x; x')$ is continuous in x . [Assumption 1.2](#) and [Assumption 1.3](#) ensure the existence of lower and upper dominance regions, respectively. Then, for any x' , the Intermediate Value Theorem implies the existence of a unique $x_i^{bench}(x')$ such that $\Delta^{bench}(x_i^{bench}(x'); x') = 0$. Since $\Delta^{bench}(x; x')$ is strictly increasing in x , investor i chooses action 2 over 0 if and only if $x \geq x_i^{bench}(x')$. Therefore, by the definition of the threshold $x'_i(x', x'')$, we have $x'_i(x', x'') \leq x_i^{bench}(x')$. Moreover, from [Proposition 2](#), there is a unique x^* such that $\Delta^{bench}(x_i^{bench}(x^*); x^*) = 0$ and

$$x_i^{bench}(x^*) = x^*. \quad (48)$$

Combining the above two observations, we have

$$x'_i(x', x'') = \min\{x_i^l(x'), x_i^{bench}(x')\}.$$

Thus, from (46), x' is the lower cutoff in a symmetric monotone equilibrium if and only if

$$x' = \min\{x_i^l(x'), x_i^{bench}(x')\}. \quad (49)$$

Thus, from (47) and (48), x' is the lower cutoff in a symmetric monotone equilibrium if and only if (i.e., (49) becomes)

$$x' = \min\{\min\{x \text{ s.t. } \Delta^1(x; x') \geq 0\}, x^*\}.$$

Next, consider the upper cutoff $x_i''(x', x'')$. Because $\pi(K(A; x'), A)$ is (strictly) increasing in A , $\delta^2(K(A; x'), A)$ is also weakly increasing in A . Then, by similar arguments made above, $\Delta^2(x; x')$ is weakly increasing in x . Additionally, $\delta^2(K(A; x'), A)$ is bounded in A and $F(A|x)$ is continuous in x , which implies that $\Delta^2(x; x')$ is continuous in x . Let $x_i^u(x')$ be the minimum x that satisfies $\Delta^2(x; x') \geq 0$, when the minimum exists, and we recall that the range of feasible x is $[\underline{A} - \sigma, \overline{A} + \sigma]$. If the minimum does not exist, set $x_i^u(x') = \overline{A} + \sigma$. That is,

$$x_i^u(x') = \min\{\{x \text{ s.t. } \Delta^2(x; x') \geq 0\} \cup \{\overline{A} + \sigma\}\}. \quad (50)$$

We now show that

$$x_i''(x', x'') = \max\{x_i^u(x'), x_i'(x', x'')\}.$$

First, consider the case where $x_i^u(x') \geq x_i'(x', x'')$. Recall that $x_i^u(x') \leq \overline{A} + \sigma$. If $x_i^u(x') < \overline{A} + \sigma$, by definition, $x_i^u(x')$ is the minimum x that solves $\Delta^2(x; x') \geq 0$. Thus, investor i chooses action 2 over 1 if and only if $x \geq x_i^u(x')$. Then, $x_i^u(x') \geq x_i'(x', x'')$ implies that she chooses action 2 over 0 as well for $x \geq x_i^u(x')$. Thus, in this case, $x_i''(x', x'') = x_i^u(x')$ because investor i chooses action 2 over any other action if and only if $x \geq x_i^u(x')$. If $x_i^u(x') = \overline{A} + \sigma$, $\Delta^2(x; x') < 0$ for all x . That is, it is never optimal to choose action 2. Then, by convention, we set $x_i''(x', x'') = \overline{A} + \sigma = x_i^u(x')$. Therefore, when $x_i^u(x') \geq x_i'(x', x'')$, investor i chooses action 2 over any other action if and only if $x \geq x_i^u(x')$ which implies $x_i''(x', x'') = x_i^u(x')$.

Second, suppose $x_i^u(x') < x_i'(x', x'')$. The existence of the upper dominance region ([Assumption 1.3](#)) implies that $x_i'(x', x'') \leq \overline{A} + \sigma$. Thus, in this case, it must be that $x_i^u(x') < \overline{A} + \sigma$. Then, by definition, $x_i^u(x')$ is the minimum x that solves $\Delta^2(x; x') \geq 0$ which implies that investor i chooses action 2 over 1 if and only if $x \geq x_i^u(x')$. Because $\Delta^2(x; x')$ is weakly increasing in x , $\Delta^2(x_i'(x', x''); x') \geq 0$, and therefore, investor i must choose action 2 over 1 for $x \geq x_i'(x', x'')$. Recall that the investor chooses either action 2 or 1 over action 0 if and only if $x \geq x_i'(x', x'')$. Thus, she must choose action 2 over actions 1 and 0 if and only if $x \geq x_i'(x', x'')$ which implies that $x_i''(x', x'') = x_i'(x', x'')$. Combining these two observations,

we get

$$x''_i(x', x'') = \max\{x''_i(x'), x'_i(x', x'')\}. \quad (51)$$

From equations (46) and (50), it follows that x'' is the upper cutoff in a symmetric monotone equilibrium if and only if

$$x'' = \max\{\min\{\{x \text{ s.t. } \Delta^2(x; x') \geq 0\} \cup \{\bar{A} + \sigma\}\}, x'_i(x', x'')\}. \quad (52)$$

Since the lower cutoff condition for the symmetric monotone equilibrium gives $x'_i(x', x'') = x'$, x'' is the upper cutoff in a symmetric monotone equilibrium if and only if

$$x'' = \max\{\min\{\{x \text{ s.t. } \Delta^2(x; x') \geq 0\} \cup \{\bar{A} + \sigma\}\}, x'\}.$$

□

[Lemma 9](#) allows us to find equilibrium cutoffs. First, we look for solutions to $\Delta^1(x'; x') = 0$, and pick the minimum when there are multiple solutions, —multiple solutions may arise, because $\Delta^1(x; x')$ is *weakly* increasing. If $x' > x^*$, we set $x' = x^*$. Then, given the solution x' we have found, we do the same for x'' . We pick x'' that solves $\Delta^2(x''; x') = 0$, and pick $x'' = \bar{A} + \sigma$ when there is no solution. Finally, if $x'' < x'$, then we set $x'' = x'$.

Next, we compute the cutoffs. First, we compute the lower cutoff x' . Consider the two cases:

Case 1 $s > \frac{\delta}{(1-\alpha)^2 \bar{K}} \left[\log \left(\frac{\bar{K} + \underline{K}}{\underline{K}} \right) - \frac{\bar{K}}{\bar{K} + \underline{K}} \right] + \sigma = x^* - (A_l - \sigma)$

Assumption 2 ensures that the set $\min\{x \text{ s.t. } \Delta^1(x; x') \geq 0\}$ is non-empty.

Note that $s > x^* - (A_l - \sigma) \implies x' < A_0(x', \sigma) - \sigma$. We can show this by first showing that $x' \geq A_0(x', \sigma) - \sigma \implies s \leq x^* - (A_l - \sigma)$. To prove this, from equation (45), x' solves

$$\begin{aligned} 0 &= E[\pi(K(A; x'), A)|x_i = x'] - t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] + s \\ s &= t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] - E[\pi(K(A; x'), A)|x_i = x'] \\ s &= t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] - \left(x' - \left(\frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2 \bar{K}} \log \left(\frac{\bar{K} + \underline{K}}{\underline{K}} \right) \right) \right) \\ s &= t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] - (x' - x^*) \\ s &\leq t \cdot E[\max\{A - A_l, 0\}|x_i = x'] - (x' - x^*) \leq (x' + \sigma - A_l) - (x' - x^*) = x^* - (A_l - \sigma). \end{aligned}$$

The last inequality follows because $x' \geq A_0(x', \sigma) - \sigma \implies x' \geq A_l - \sigma$. Then, taking the contrapositive gives $s > x^* - (A_l - \sigma) \implies x' < A_0(x', \sigma) - \sigma$.

When $x' < A_0(x', \sigma) - \sigma$, the posterior support $[x' - \sigma, x' + \sigma]$ lies below $A_0(x', \sigma) = A_l$, therefore, $E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] = 0$. Then, because the marginal investor gets a payoff of zero, at $x_i = x'$,

$$\begin{aligned} 0 &= E[\pi(K(A; x'), A)|x_i = x'] - t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] + s \\ 0 &= E[\pi(K(A; x'), A)|x_i = x'] + s \\ x' &= \frac{r}{\alpha} + \frac{\delta}{(1-\alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - s = x^* - s. \end{aligned}$$

Case 2: $s \leq \frac{\delta}{(1-\alpha)^2 \bar{K}} \left[\log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{\bar{K}}{\bar{K} + \underline{K}} \right] + \sigma = x^* - (A_l - \sigma)$

In this case, $x' \geq A_0(x', \sigma) - \sigma$. We can show this by contradiction. Suppose $x' < A_0(x', \sigma) - \sigma$. From equation (15), $A_0(x', \sigma) = A_l$ when $x' < A_0(x', \sigma) - \sigma$ and, therefore, $E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] = 0$. Then, because x' solves $E[\pi(K(A; x'), A)|x_i = x'] - t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] + s = 0$, $s = x^* - x'$. Because $A_0(x', \sigma) = A_l$, $x' < A_0(x', \sigma) - \sigma = A_l - \sigma$. Then, $x^* - s < A_l - \sigma \implies s > x^* - (A_l - \sigma)$ which is a contradiction. Therefore, $s \leq x^* - (A_l - \sigma) \implies x' \geq A_0(x', \sigma) - \sigma$.

Also, $x' < A_0(x', \sigma) + \sigma$ because $x' \leq x^* \leq A_h$. Therefore, in this case, $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$. Next, the lower cutoff x' is bounded at x^* . When $x' < x^*$, the lower cutoff is given by

$$\begin{aligned}
E[\pi(1, K(A; x'), A)|x_i = x'] &= E[\pi(K(A; x'), A)|x_i = x'] - t \cdot E[\max\{\pi(K(A; x'), A), 0\}|x_i = x'] + s \\
0 &= E\left[\left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right)\middle| x_i = x'\right] \\
&\quad - t \cdot E\left[\max\left\{\left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right), 0\right\}\middle| x_i = x'\right] + s \\
0 &= \int_{x'-\sigma}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right) f(A|x_i = x') dA \\
&\quad - t \left\{ \int_{x'-\sigma}^{A_0(x', \sigma)} 0 \cdot f(A|x_i = x') dA + \int_{A_0(x', \sigma)}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right) f(A|x_i = x') dA \right\} + s \\
0 &= \int_{x'-\sigma}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right) f(A|x_i = x') dA \\
&\quad - t \cdot \int_{A_0(x', \sigma)}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right) f(A|x_i = x') dA + s.
\end{aligned}$$

Solving the above equation,

$$\begin{aligned}
x' - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2 \bar{K}} \log\left(\frac{\underline{K} + \bar{K}}{\underline{K}}\right) - \frac{t}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} \right. \\
\left. - \frac{2\sigma\delta}{(1-\alpha)^2 \bar{K}} \log\left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2}\left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma}\right)\right)}\right) \right] = -s \\
x' - \frac{t}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} \right. \\
\left. - \frac{2\sigma\delta}{(1-\alpha)^2 \bar{K}} \log\left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2}\left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma}\right)\right)}\right) \right] = x^* - s.
\end{aligned} \tag{53}$$

The lower cutoff x' solves $x' = T(x')$ where

$$\begin{aligned}
T(x') &= \frac{t}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} \right. \\
&\quad \left. - \frac{2\sigma\delta}{(1-\alpha)^2 \bar{K}} \log\left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2}\left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma}\right)\right)}\right) \right] + x^* - s,
\end{aligned}$$

and $x' \in X' := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x < A_0(x, \sigma) + \sigma\}$. $T(x)$ is differentiable over $X' := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x < A_0(x, \sigma) + \sigma\}$. Then, the slope $T'(x)$ is

$$\begin{aligned}
T'(x) &= \frac{t}{2\sigma} \left[x + \sigma - A_0(x, \sigma) A'_0(x, \sigma) - \frac{r(1 - A'_0(x, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1 - \alpha)^2} \frac{1}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma}\right)\right)} \frac{1 - A'_0(x, \sigma)}{2\sigma} \right] \\
&= \frac{t}{2\sigma} \left[x + \sigma - A'_0(x, \sigma) \left(A_0(x, \sigma) - \frac{r}{\alpha} - \frac{\delta}{(1 - \alpha)^2} \frac{1}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma}\right)\right)} \right) \right. \\
&\quad \left. - \frac{r}{\alpha} - \frac{\delta}{(1 - \alpha)^2} \frac{1}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x - A_0(x, \sigma)}{\sigma}\right)\right)} \right] \\
&= \frac{t}{2\sigma} \left[x + \sigma - 0 - A_0(x, \sigma) \right] \quad \{from \text{ equation (15)}\} \\
&= \frac{t}{2\sigma} (x - A_0(x, \sigma) + \sigma).
\end{aligned}$$

Since $x \in X' := \{x \in \mathbb{R} : A_0(x, \sigma) - \sigma \leq x < A_0(x, \sigma) + \sigma\}$ and $t \in [0, 1]$, $T'(x) \in [0, 1] \implies T'(x) < 1$. It follows that $T(x)$ admits a unique fixed point. Thus, when $x' < x^*$, x' uniquely solves equation (53).

Since the cutoff with program cannot be greater than the benchmark cutoff x^* , $x' \leq x^*$ (follows from equation (45)), the lower cutoff in equilibrium when $s \leq \frac{\delta}{(1 - \alpha)^2 \bar{K}} \left[\log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{\bar{K}}{\bar{K} + \underline{K}} \right] + \sigma$ is

$$x' = \min\{\tilde{x}(s, t), x^*\},$$

where $\tilde{x}$ uniquely solves,

$$\tilde{x} = \frac{t}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1 - \alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + K(A_0(\tilde{x}, \sigma); \tilde{x})} \right) \right] + x^* - s. \quad (54)$$

Thus, there is a unique x' in the equilibrium. Finally, note that for $s = x^* - A_l + \sigma$, $\tilde{x} = x^* - s$ solves equation (54) implying $x'(s, t) = x^* - s$ in this case.

Next, we compute the upper cutoff x'' . When $s \geq t(\bar{A} - A_l) = t\pi(\bar{K}, \bar{A})$, $t\pi(\bar{K}, \bar{A}) - s = \pi(2, \bar{K}, \bar{A}) - \pi(1, \bar{K}, \bar{A}) \leq 0$. Therefore, the upper cutoff is $x'' = \bar{A} + \sigma$ because any investor that invests participates in the program.

For $\frac{s}{t} < \pi(\bar{K}, \bar{A}) = \bar{A} - A_l$, because $\pi(1, K(\bar{A}, x'), \bar{A}) - \pi(2, K(\bar{A}, x'), \bar{A}) < 0$, $x'' < \bar{A} + \sigma$.

Consider the indifference condition

$$\frac{s}{t} = E[\max\{\pi(K(A; x'), A), 0\} | x_i]. \quad (55)$$

Note that $E[\max\{\pi(K(A; x'), A), 0\} | x_i] = E\left[\max\left\{\left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right), 0\right\} \middle| x_i\right] \geq 0$ and is weakly increasing in x_i . Also, $E\left[\max\left\{\left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right), 0\right\} \middle| x_i\right]$ is strictly increasing in x_i when the conditional expectation is positive. Consider the following cases:

- If $\frac{s}{t} > 0$,

In this case, $E\left[\max\left\{\left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right), 0\right\} \middle| x_i\right] > 0$ and because it is strictly increasing, there is a unique $x^\diamond$ that solves,

$$\frac{s}{t} = E\left[\max\left\{\left(A - \frac{\delta}{(1-\alpha)^2} \frac{1}{(\underline{K} + K(A; x'))} - \frac{r}{\alpha}\right), 0\right\} \middle| x_i = x^\diamond\right].$$

- If $s/t = 0$,

$x' = x^*$ and equation (55) is satisfied for all $x_i \leq x^* - \sigma$. Then, by the tie breaking rule (we choose the lowest signal), $x^\diamond = \underline{A} - \sigma$.

Finally, because $x'' \geq x'$, the unique upper cutoff is given by

$$x'' = \max\{x', x^\diamond(s, t)\}.$$

□

Proof of Proposition 11. The proof follows a similar approach to the proof of Proposition 10. The monotone best-response property and the cutoff characterization (analogous to Lemma 8 and equation (45)) apply here as well with the only difference being the payoff differences that are replaced by the GPLS analog. Therefore, the equilibrium is pinned down by a cutoff pair (x', x'') . To avoid repetition, we do not restate these intermediate steps and only focus on characterizing the two cutoffs x' and x'' .

First, consider the lower cutoff x' .

Case 1: $\frac{s}{1-t^-} > \frac{\delta}{(1-\alpha)^2 \bar{K}} \left[\log\left(\frac{\underline{K} + \bar{K}}{\underline{K}}\right) - \frac{\bar{K}}{\bar{K} + \underline{K}} \right] + \sigma = x^* - A_l + \sigma$

When $\frac{s}{1-t^-} > x^* - A_l + \sigma$, $x' < A_0(x', \sigma) - \sigma$. The arguments for this are similar to that in the proposition for the GPS. Let $t^+ = t^- + \gamma$, then for a GPLS program (s, t^+, t^-) , x' solves

$$s = t^+ E[\max\{\pi(K(A; x'), A), 0\} | x_i = x'] + t^- E[\min\{\pi(K(A; x'), A), 0\} | x_i = x'] \\ - E[\pi(K(A; x'), A) | x_i = x']$$

$$s = \gamma E[\max\{\pi(K(A; x'), A), 0\} | x_i = x'] - (1 - t^-) E[\pi(K(A; x'), A) | x_i = x'] \quad \{\because \gamma = t^+ - t^-\} \\ \frac{s}{1 - t^-} = \frac{\gamma}{1 - t^-} E[\max\{\pi(K(A; x'), A), 0\} | x_i = x'] - E[\pi(K(A; x'), A) | x_i = x'] \quad (56)$$

$$\text{Then, } \frac{s}{1 - t^-} \leq \frac{\gamma}{1 - t^-} E[\max\{A - A_l, 0\} | x_i = x'] - (x' - x^*) \leq x' - A_l + \sigma - (x' - x^*) = x^* - A_l + \sigma.$$

The last inequality follows because $x' \geq A_0(x', \sigma) - \sigma \implies x' \geq A_l - \sigma$. Therefore, if $x' \geq A_0(x', \sigma) - \sigma$, $\frac{s}{1 - t^-} \leq x^* - A_l + \sigma$. The contrapositive gives us $\frac{s}{1 - t^-} > x^* - A_l + \sigma \implies x' < A_0(x', \sigma) - \sigma$.

When $x' < A_0(x', \sigma) - \sigma$, $A_0(x', \sigma) = A_l$ and therefore, $E[\max\{\pi(K(A; x'), A), 0\} | x_i = x'] = 0$ and $E[\min\{\pi(K(A; x'), A), 0\} | x_i = x'] = E[\pi(K(A; x'), A) | x_i = x']$. The lower cutoff is the signal at which her payoff from investing while participating in the program is 0, i.e., $E[\pi_{gpls}(1, K(A; x'), A) | x_i = x'] = 0$. Therefore,

$$0 = E[\pi(K(A; x'), A) | x'] - t^+ \cdot E[\max\{\pi(K(A; x'), A), 0\} | x'] - t^- \cdot E[\min\{\pi(K(A; x'), A), 0\} | x'] + s \\ 0 = (1 - t^-) E[\pi(K(A; x'), A) | x'] + s \quad \{\because x' < A_0(x', \sigma) - \sigma\} \\ x' = \frac{r}{\alpha} + \frac{\delta}{(1 - \alpha)^2 \bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{s}{1 - t^-}.$$

$$\textbf{Case 2: } \frac{s}{1 - t^-} \leq \frac{\delta}{(1 - \alpha)^2 \bar{K}} \left[\log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) - \frac{\bar{K}}{\bar{K} + \underline{K}} \right] + \sigma = x^* - A_l + \sigma$$

Using the same arguments as in the proof of [Proposition 10](#), in this case, $x' \in [A_0(x', \sigma) - \sigma, A_0(x', \sigma) + \sigma]$. The lower cutoff x' solves $E[\pi_{gpls}(1, K(A; x'), A) | x_i = x'] = 0$. Thus,

$$\begin{aligned}
0 &= E[\pi(K(A; x'), A)|x'] - t^+ \cdot E[\max\{\pi(K(A; x'), A), 0\}|x'] - t^- \cdot E[\min\{\pi(K(A; x'), A), 0\}|x'] + s \\
0 &= \int_{x'-\sigma}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha} \right) f(A|x_i = x') dA \\
&\quad - t^+ \cdot \int_{A_0(x', \sigma)}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha} \right) f(A|x_i = x') dA \\
&\quad - t^- \cdot \int_{x'-\sigma}^{A_0(x', \sigma)} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha} \right) f(A|x_i = x') dA + s \\
x' &= t^+ \cdot \int_{A_0(x', \sigma)}^{x'+\sigma} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha} \right) f(A|x_i = x') dA \\
&\quad + t^- \cdot \int_{x'-\sigma}^{A_0(x', \sigma)} \left(A - \frac{\delta}{(1-\alpha)^2(\underline{K} + K(A; x'))} - \frac{r}{\alpha} \right) f(A|x_i = x') dA + x^* - s \\
x' &= \frac{t^+}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right) \right] \\
&\quad + \frac{t^-}{2\sigma} \left[\frac{A_0(x', \sigma)^2 - (x' - \sigma)^2}{2} - \frac{r(A_0(x', \sigma) - x' + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)}{\underline{K}} \right) \right] \\
&\quad + x^* - s.
\end{aligned}$$

The RHS of the above equation has a slope less than 1. To prove this, let $t^+ = t^- + \gamma$. Then,

$$\begin{aligned}
x' &= \frac{\gamma}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right) \right] \\
&\quad + \frac{t^-}{2\sigma} \left[\frac{(x' + \sigma)^2 - (x' - \sigma)^2}{2} - \frac{r(x' + \sigma - x' + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K}} \right) \right] + x^* - s \\
x' &= \frac{\gamma}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right) \right] \\
&\quad + t^- (x' - x^*) + x^* - s \\
x' &= \frac{\gamma}{1-t^-} \frac{1}{2\sigma} \left[\frac{(x' + \sigma)^2 - A_0(x', \sigma)^2}{2} - \frac{r(x' + \sigma - A_0(x', \sigma))}{\alpha} \right. \\
&\quad \left. - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right) \right] + x^* - \frac{s}{1-t^-}.
\end{aligned}$$

Taking derivative of the RHS w.r.t. x' ,

$$\begin{aligned}
& \left(\frac{\gamma}{1-t^-} \right) \frac{1}{2\sigma} \left[x' + \sigma - A_0(x', \sigma) A'_0(x', \sigma) - \frac{r}{\alpha} + \frac{r}{\alpha} A'_0(x', \sigma) - \frac{\delta}{(1-\alpha)^2} \frac{1 - A'_0(x', \sigma)}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right] \\
&= \left(\frac{\gamma}{1-t^-} \right) \frac{1}{2\sigma} \left[x' + \sigma - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right. \\
&\quad \left. - A'_0(x', \sigma) \left(A_0(x', \sigma) - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right) \right] \\
&= \left(\frac{\gamma}{1-t^-} \right) \frac{1}{2\sigma} \left[x' + \sigma - \frac{r}{\alpha} - \frac{\delta}{(1-\alpha)^2} \frac{1}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{x' - A_0(x', \sigma)}{\sigma} \right) \right)} \right] \\
&= \left(\frac{\gamma}{1-t^-} \right) \frac{1}{2\sigma} \left[x' + \sigma - A_0(x', \sigma) \right] < 1.
\end{aligned}$$

The last inequality follows because $\left(\frac{\gamma}{1-t^-} \right) \leq 1$ and $(x' - A_0(x', \sigma) + \sigma) \in [0, 2\sigma)$. Thus, there exists a unique x' . Moreover, because x' is bounded above at x^* , the lower cutoff is

$$x'(s, t^+, t^-) = \min\{\tilde{x}(s, t^+, t^-), x^*\},$$

where $\tilde{x}$ uniquely solves

$$\begin{aligned}
\tilde{x} &= \frac{t^+}{2\sigma} \left[\frac{(\tilde{x} + \sigma)^2 - A_0(\tilde{x}, \sigma)^2}{2} - \frac{r(\tilde{x} + \sigma - A_0(\tilde{x}, \sigma))}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \bar{K}}{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)} \right) \right] \\
&\quad + \frac{t^-}{2\sigma} \left[\frac{A_0(\tilde{x}, \sigma)^2 - (\tilde{x} - \sigma)^2}{2} - \frac{r(A_0(\tilde{x}, \sigma) - \tilde{x} + \sigma)}{\alpha} - \frac{2\sigma\delta}{(1-\alpha)^2\bar{K}} \log \left(\frac{\underline{K} + \frac{\bar{K}}{2} \left(1 - \left(\frac{\tilde{x} - A_0(\tilde{x}, \sigma)}{\sigma} \right) \right)}{\underline{K}} \right) \right] \\
&\quad + x^* - s.
\end{aligned}$$

Finally, note that for $\frac{s}{1-t^-} = x^* - A_l + \sigma$, $\tilde{x} = x^* - \frac{s}{1-t^-}$ solves the above equation implying $x'(s, t^+, t^-) = x^* - \frac{s}{1-t^-}$ in this case.

Next, for the upper cutoff x'' , when $s/t^+ \geq \bar{A} - A_l = \pi(\bar{K}, \bar{A})$, $\pi_{gpls}(2, \bar{K}, \bar{A}) - \pi_{gpls}(1, \bar{K}, \bar{A}) \leq 0$ implying that $x'' = \bar{A} + \sigma$. When $s/t^+ < \bar{A} - A_l$, $x'' < \bar{A} + \sigma$ and x'' solves

$$s = t^+ \cdot E[\max\{\pi(K(A; x'), A), 0\} | x_i = x] + t^- \cdot E[\min\{\pi(K(A; x'), A), 0\} | x_i = x]. \quad (57)$$

Because $E[\pi(K(A; x'), A) | x_i]$ is strictly increasing in x_i , $E[\max\{\pi(K(A; x'), A), 0\} | x_i = x]$ and $E[\min\{\pi(K(A; x'), A), 0\} | x_i = x]$ are weakly increasing in x , and the RHS of (57) is weakly increasing in x when $s/t^+ \leq \pi(\bar{K}, \bar{A})$. At $x_i = \bar{A} + \sigma$, the investor strictly prefers non-participation: at sufficiently low signals, participation dominates (or the investor doesn't invest at all). Combined with the weak monotonicity of the RHS of (57), the intermediate

value theorem gives the existence of $x^\diamond$. Let $x^\diamond(s, t^+, t^-)$ denote the minimum x that solves equation (57). Furthermore, $x'' \geq x'$. Therefore, the upper cutoff is

$$x'' = \max\{x', x^\diamond(s, t^+, t^-)\}. \quad \square$$